

Generation/Amplification of high
frequency-High Power signal

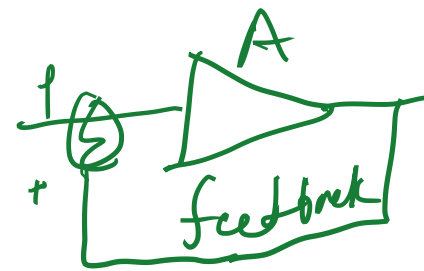
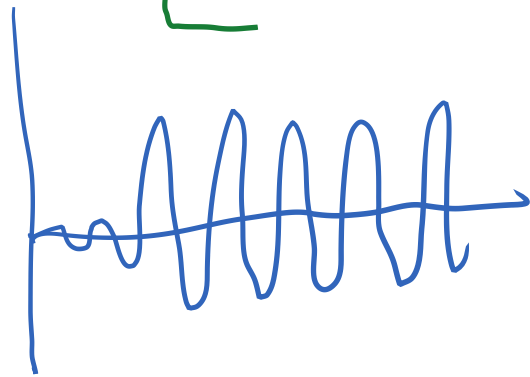
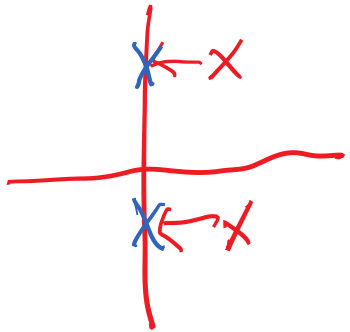
Oscillator + To generate a high frequency signal.

* $\rightarrow$ Negative resistance offered device (Active Device)

$g^2 R \rightarrow R \rightarrow +ve$ [heat dissipation/loss]
 $R \rightarrow -ve$ [heat/energy generation]

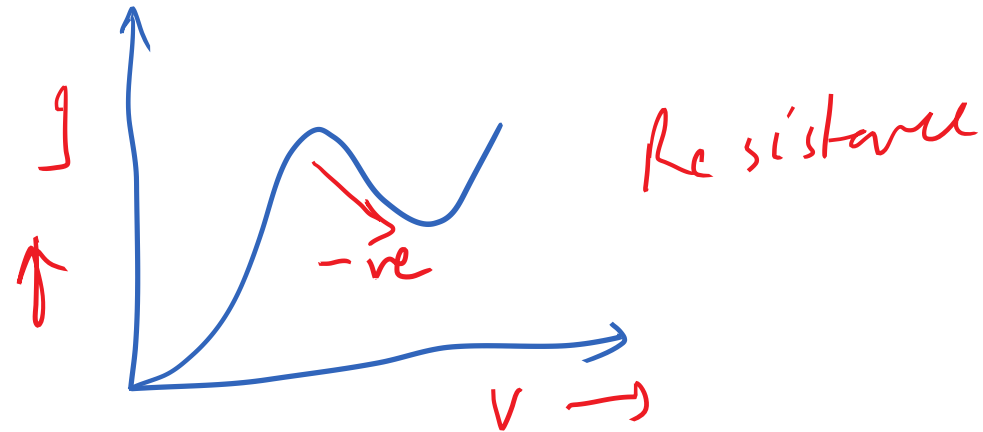
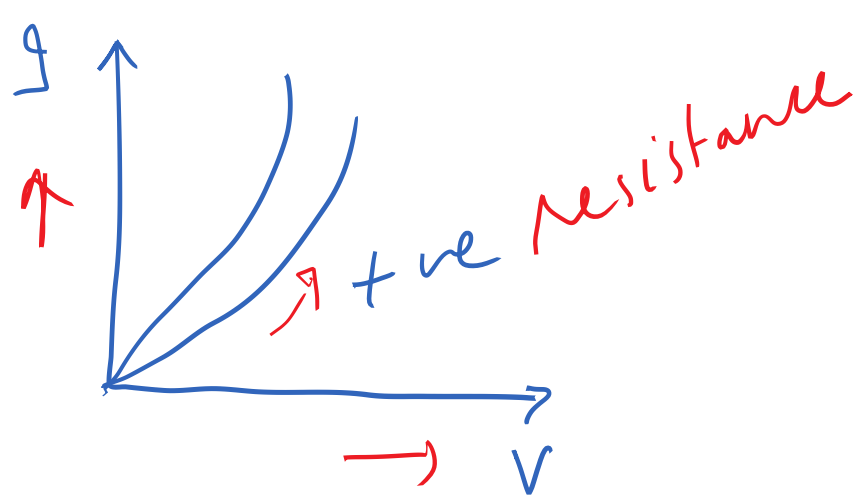
Active device

- BJT
- MOSFET

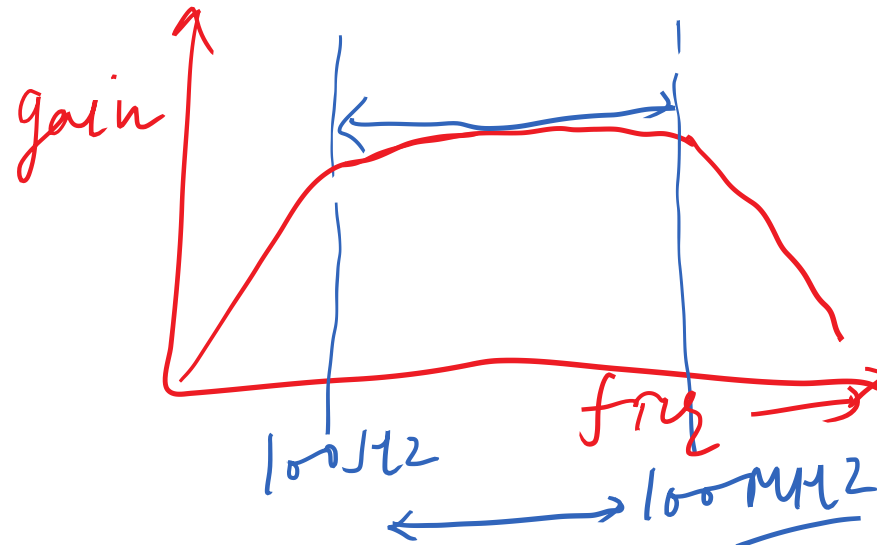
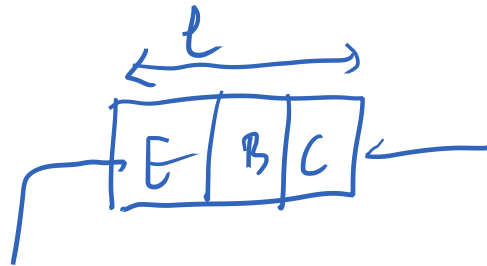


$$\left[\frac{A}{1 - AB} \right] \text{ oscillator}$$

Positive & Negative resistance device



BJT $\rightarrow$ 547/147



high $f \uparrow$
 $t \downarrow$
 $l \downarrow$

So Need devices to generate high freq. signals.

FET

- HEMT → High electron mobility Transistor
- MESFET → Metal Semiconductor field effect transistor

BJT

- HBT → Heterojunction Bipolar Transistor

3 terminal Devices

2 Terminal Devices

- Gunn Diode
- IMPATT Diode
- TRAPATT Diode
- Tunnel Diode
- BARITT Diode

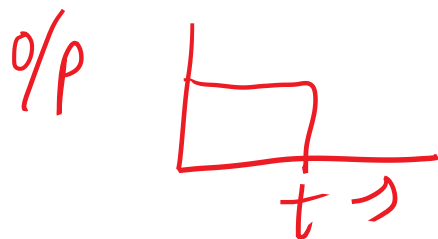
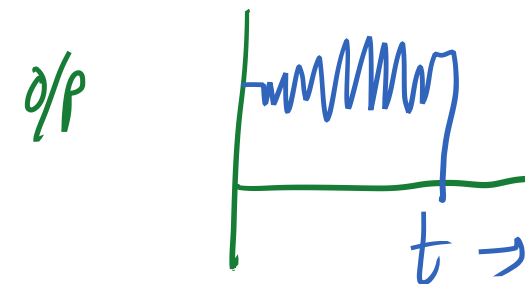
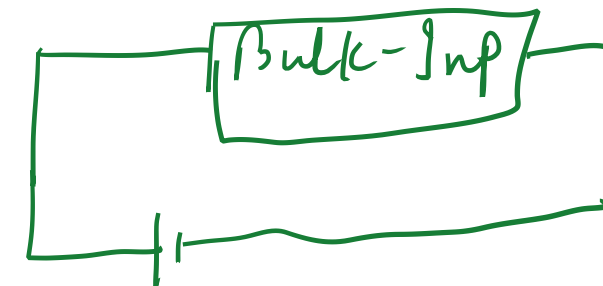
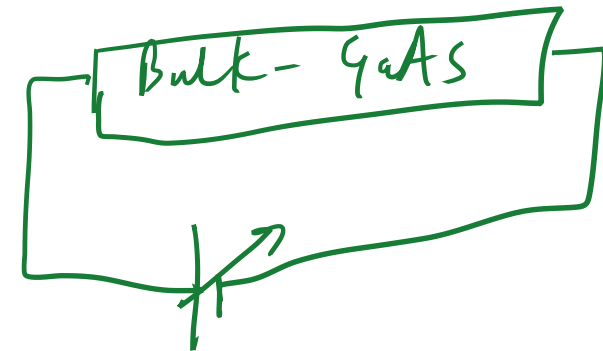
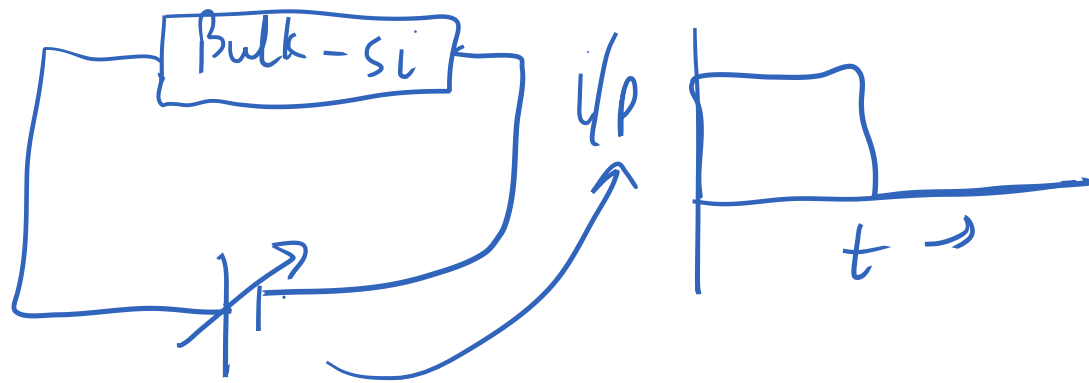
1-100W
 $f < 20 \text{ GHz}$
P-n junction

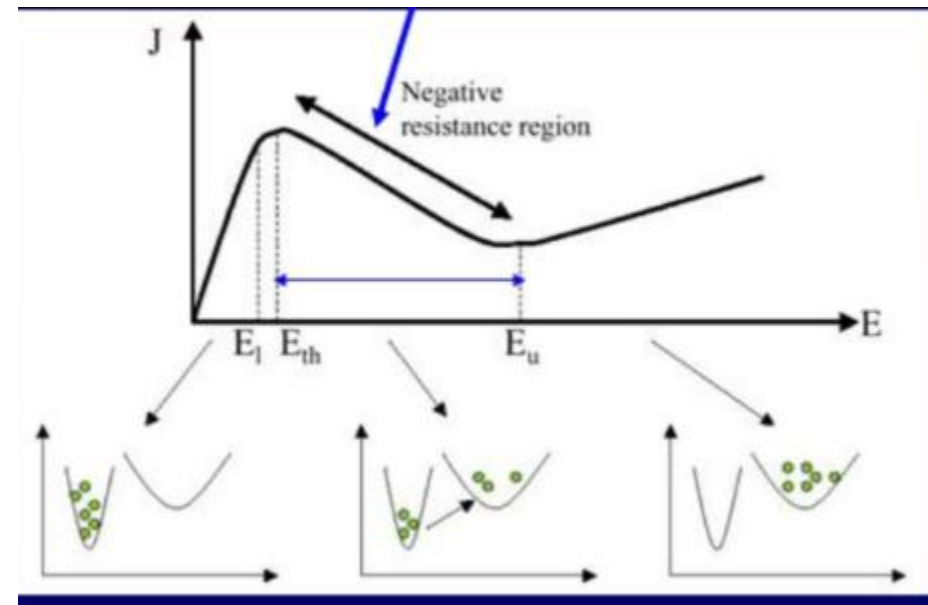
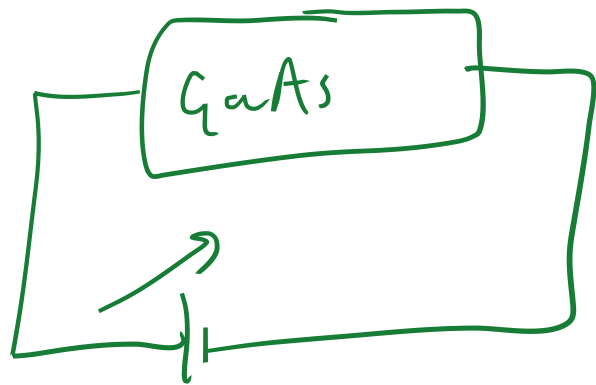
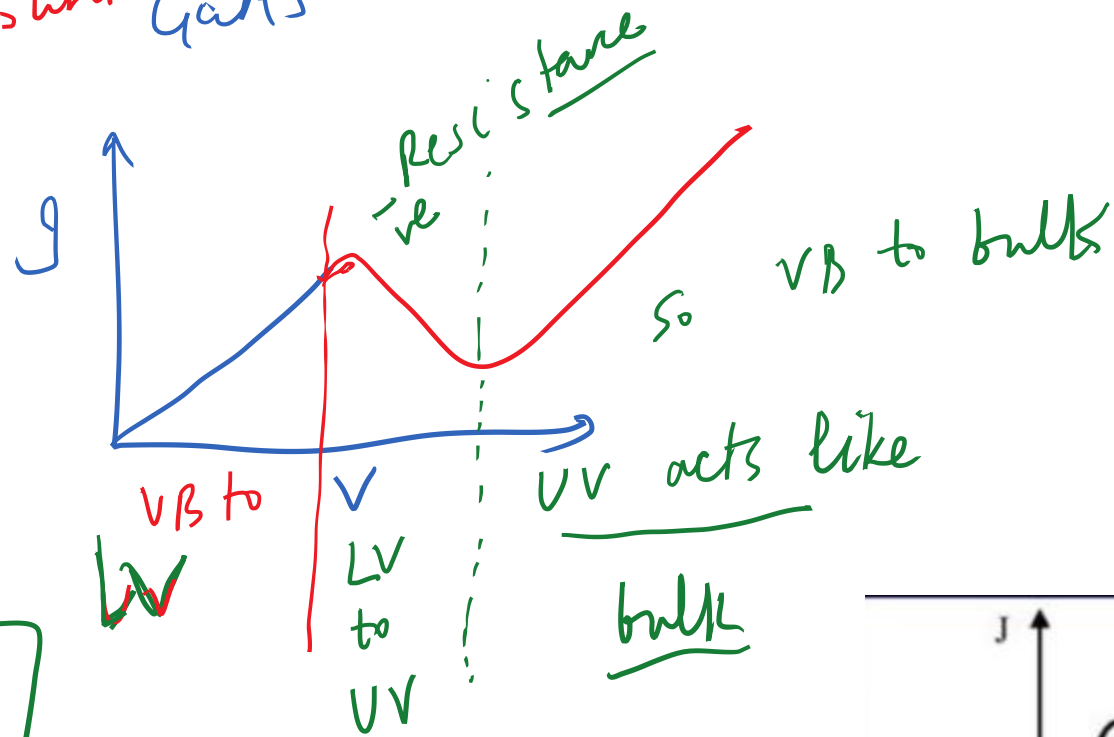
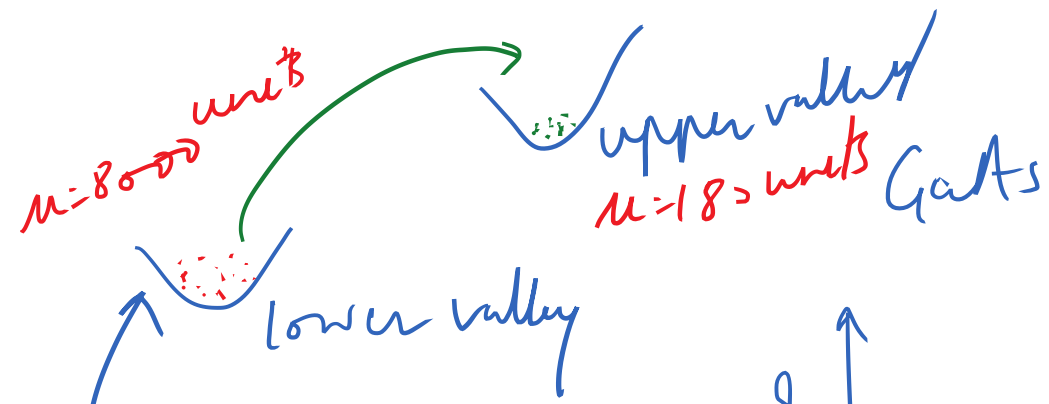
* Power generate in these devices are very small

* Smaller Size

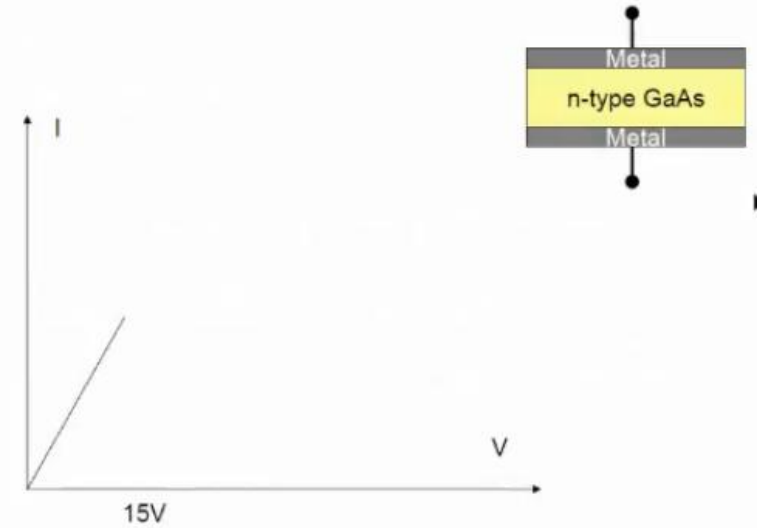
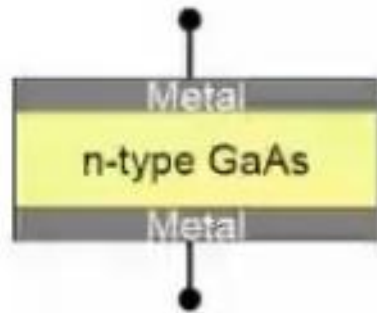
Tubes + High power Signal
GUNN Diode

Multiple conduction
band

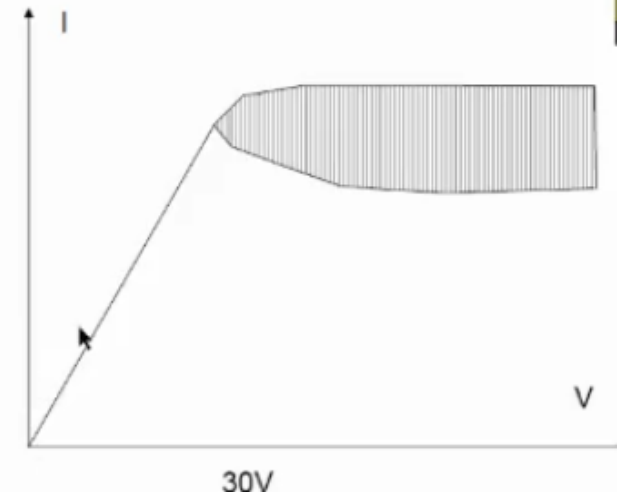


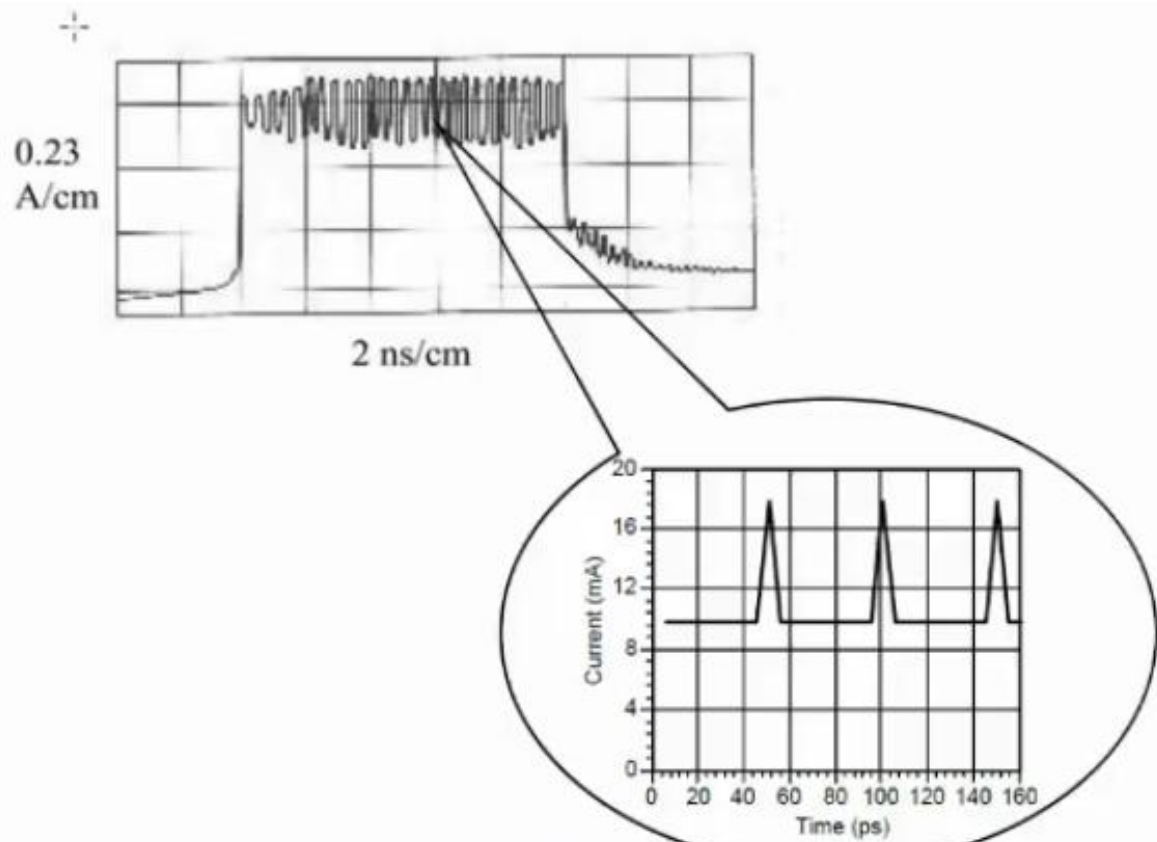


- ▶ A Gunn Diode is considered as a type of diode even though it does not contain any typical PN diode junction like the other diodes, but it consists of two electrodes.
- ▶ This diode is also called as a Transferred Electronic Device. This diode is a negative differential resistance device, which is frequently used as a low-power oscillator to generate microwaves.
- ▶ It consists of only N-type semiconductor in which electrons are the majority charge carriers. To generate short radio waves such as microwaves, it utilizes the Gunn Effect.



GaAs sample I-V characteristic in Gunn experiments



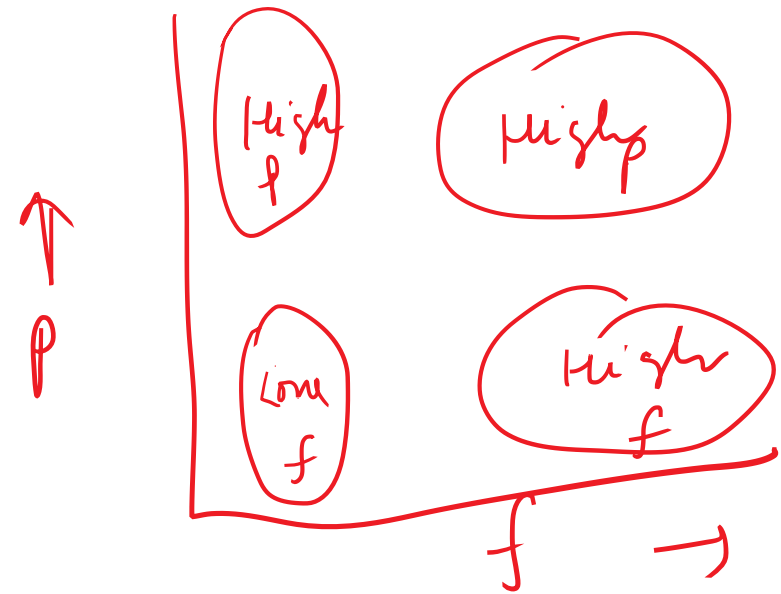


Short-pulse current waveform in Gunn experiment

* Generate - (High f , High P)

* Amplify - (High f , High P)

done power



Gunn diode $\rightarrow$ high frequency

But device size smaller

so more power

Dissipation

why semiconductor can't be used for high $P \uparrow$ high f

a) Transit time $\rightarrow$ high $f \uparrow$ transit time $\downarrow$ device size $\downarrow$ Power density

b) Power dissipation! Too high

c) Parasitic effect! In semiconductor devices

Applications: High $f \uparrow$ High $P \uparrow$

a) Ovens (microwave),

b) Weapons (Army)

c) Lasers

heating purpose

[$\rightarrow$ How to generate power $\rightarrow$ Magnetron $\uparrow$ High f High P]

High f High P

Generate

- Magnetron
- BWO → Backward wave Oscillator
- Reflex klystron
- Gyrotrons

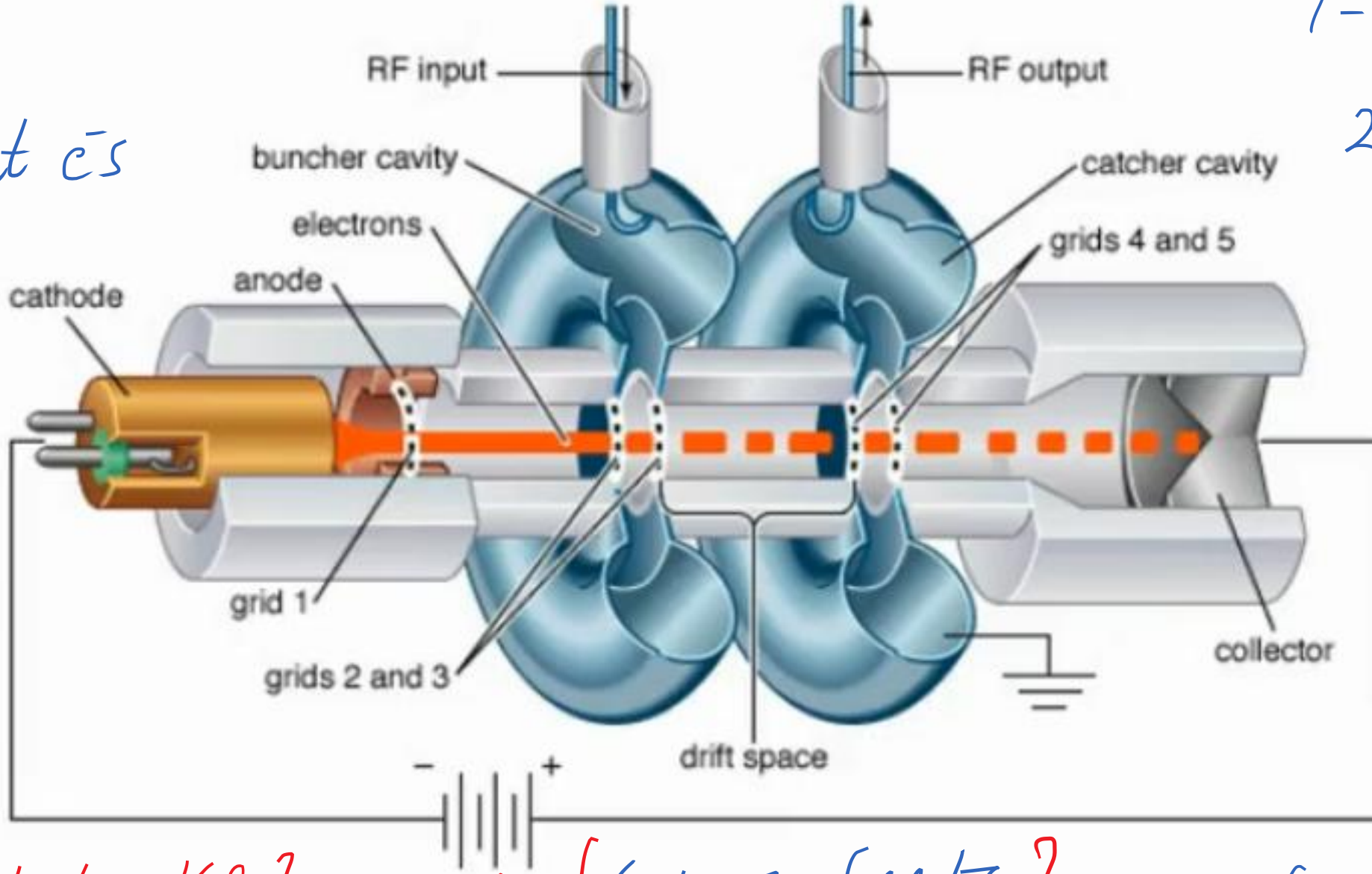
Amplify

- Klystron
- TWT

↓
Traveling Wave Tubes

Klystron

emit e^-



1-5 GHz

200 kW

500 kW

$\eta = 50\%$

$P_{out} = 500 \text{ kW}$

$P_{in} = 1 \text{ MW}$

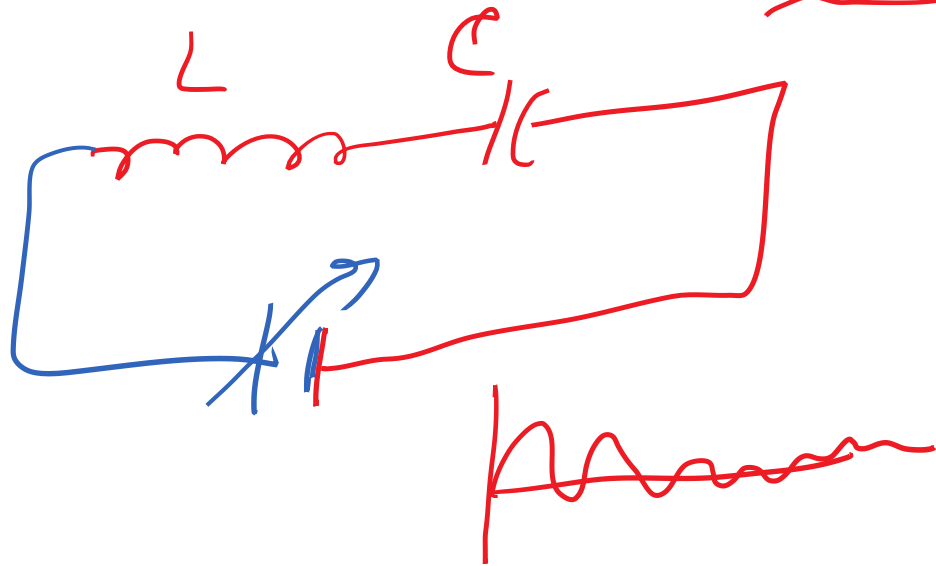
$50 \text{ kV} \rightarrow 20 \text{ A}$

$W =$
[500 to 600 kg]

$L =$ [6 to 7 feet]

Amplifier/Oscillator

↳ Inte need resonance ckt.



$$\left[f = \frac{1}{2\pi\sqrt{LC}} \right]$$

for high $f \uparrow LC \downarrow$

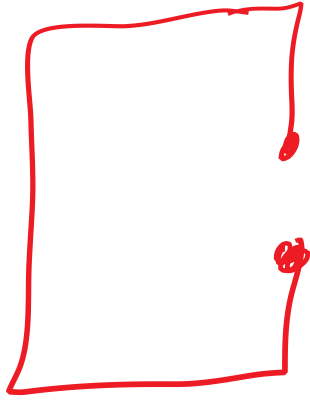
$C \downarrow$

How to reduce L ??



$L \rightarrow$ 

Reduce the number of turns.



It's a resonator with

some L & C

✓ make many



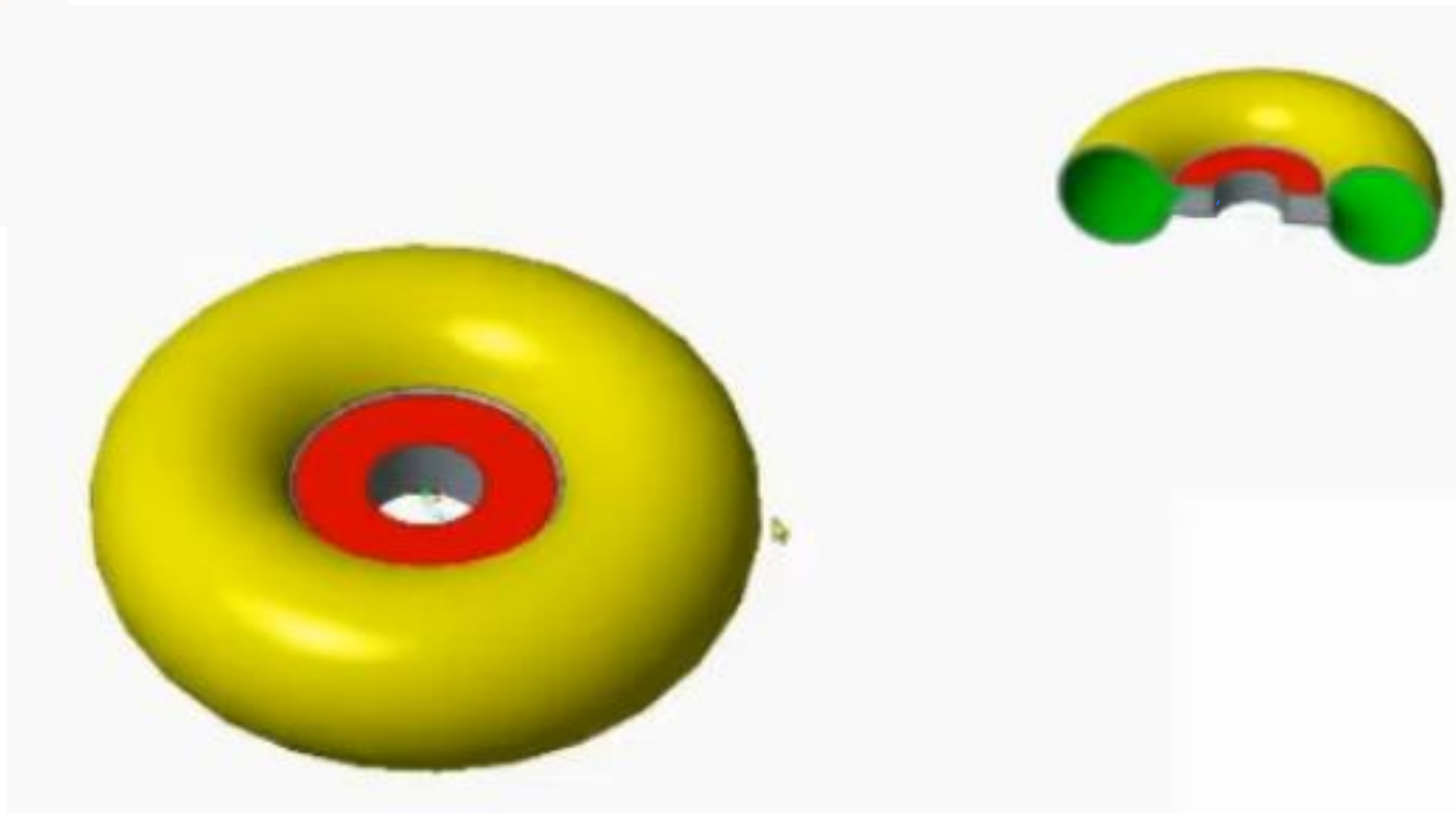
I want to increase the frequency

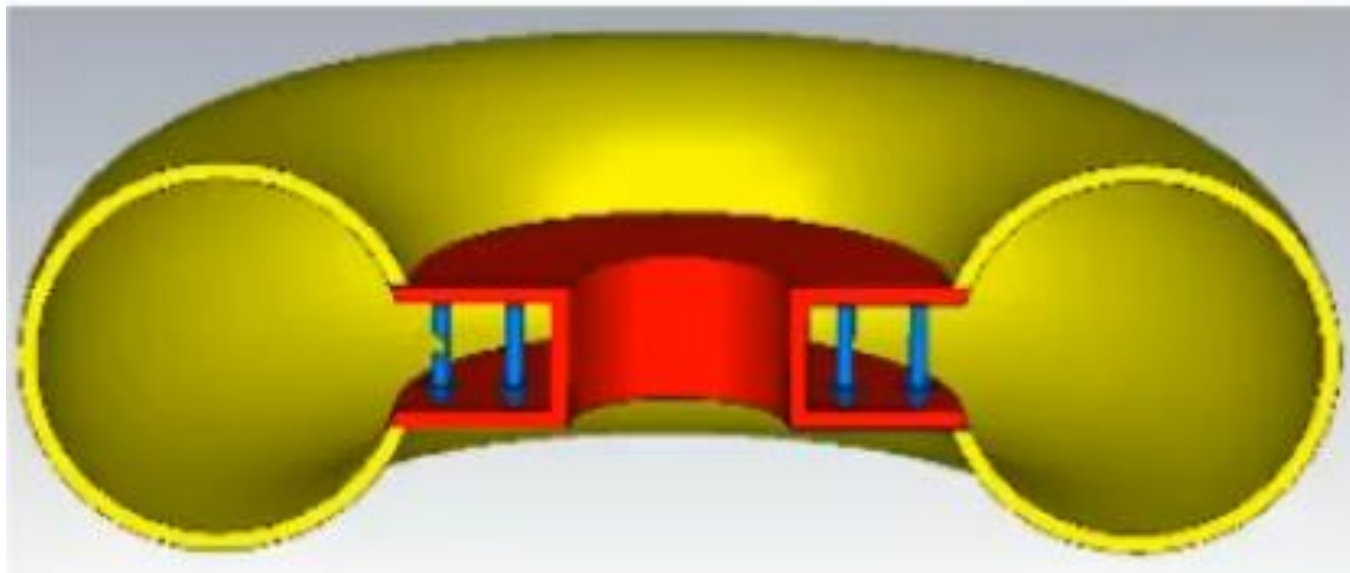
↓ Reduce L

inductances in parallel ✓



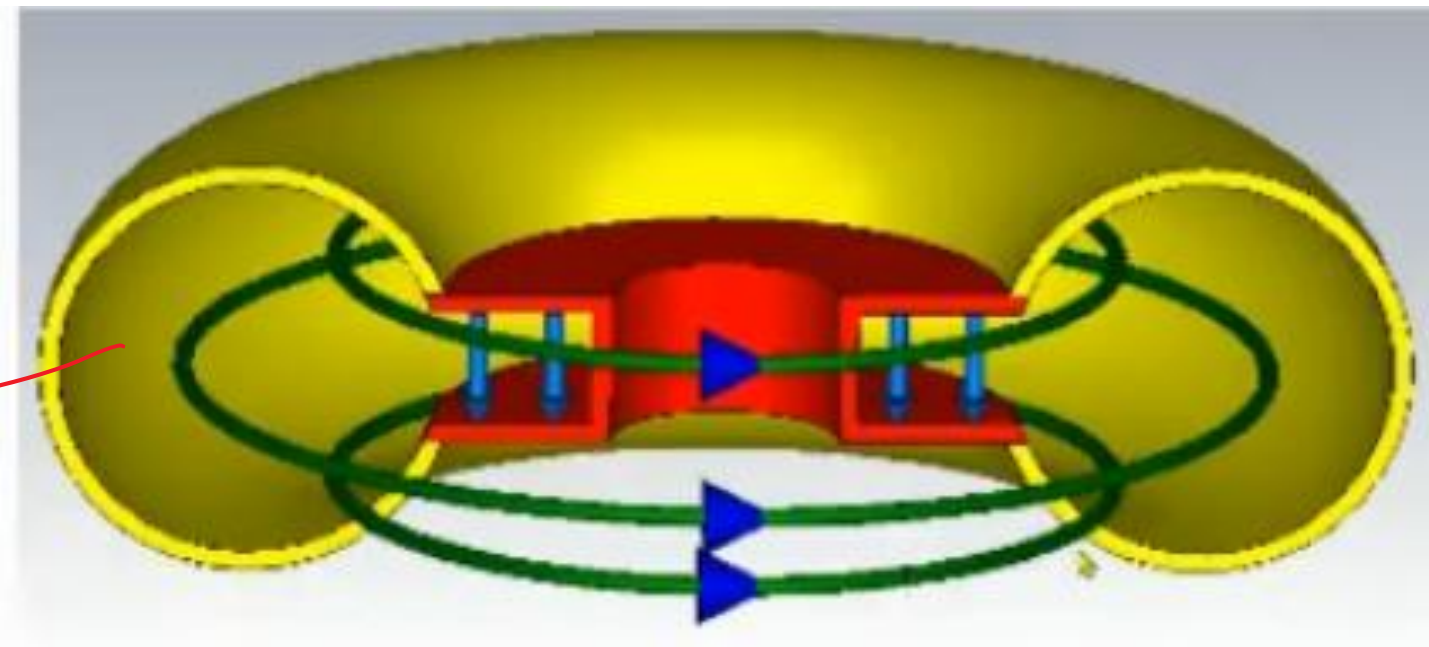
parallel inductances





Electric field

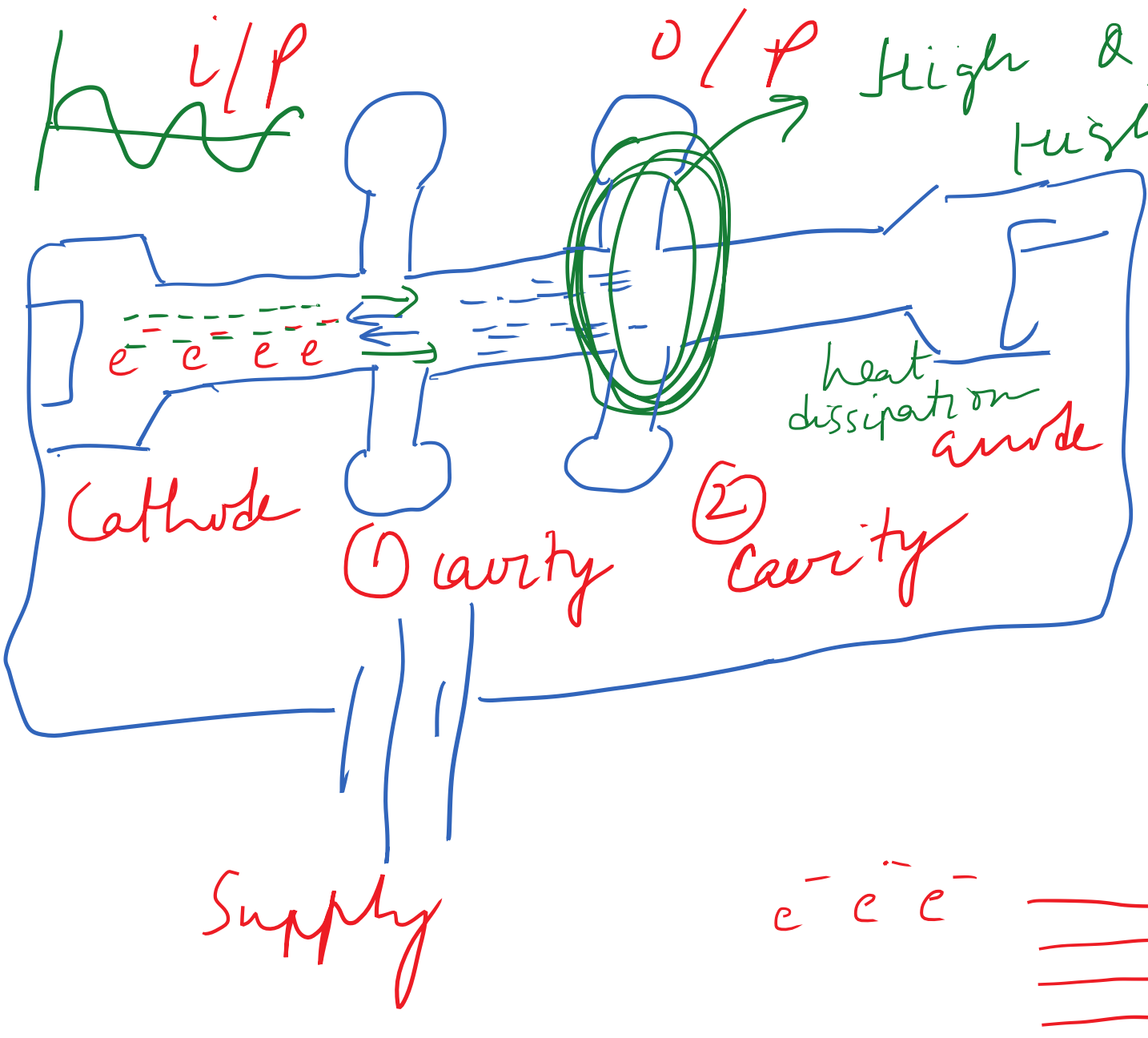
Magnetic field



i/p

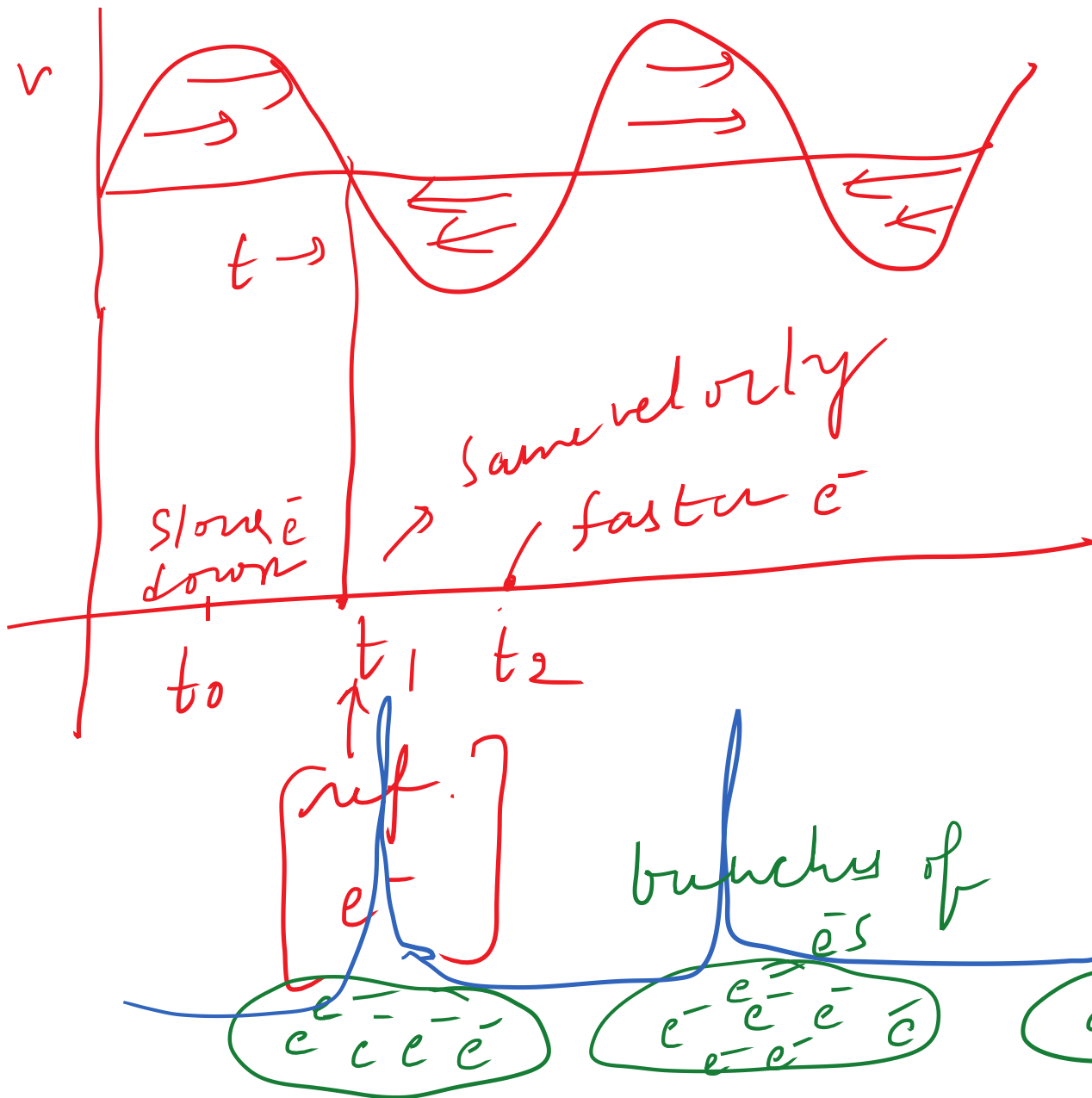
o/p

High Q resonator
high $f \rightarrow$ more current
more bunches



$e^- e^- e^-$
accelerated

$e^- e^- e^-$
deacceleroff



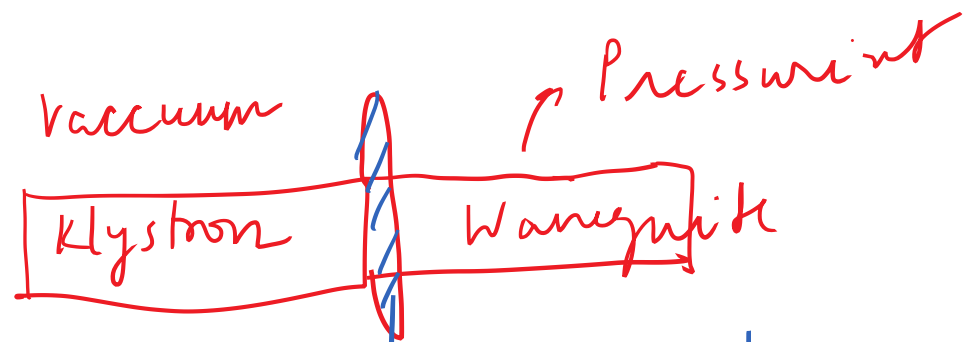
Vacuum
Tube
Structure

velocity
modulation

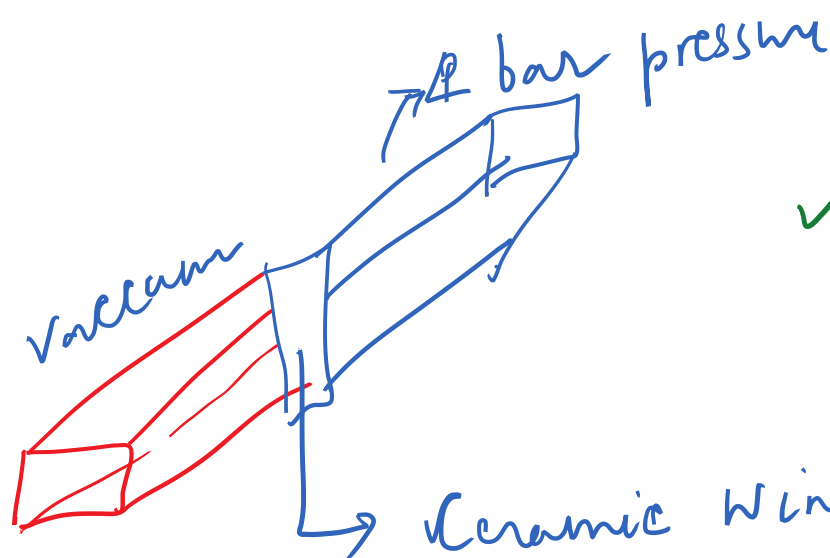
clustion
↓
klystron

klystron

works only
for one frequency.

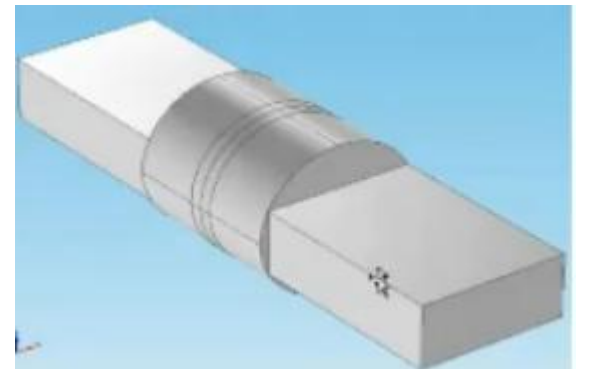


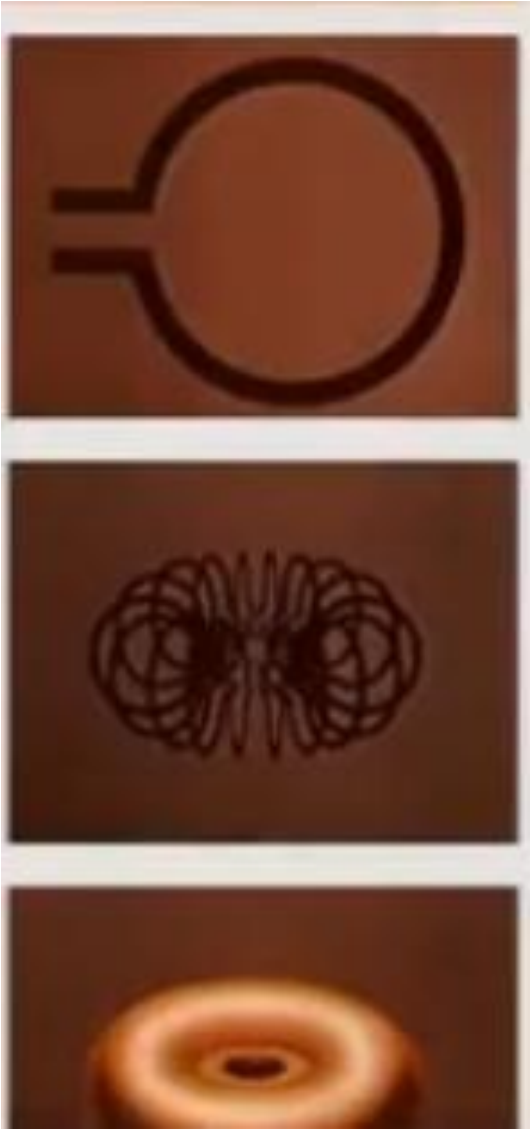
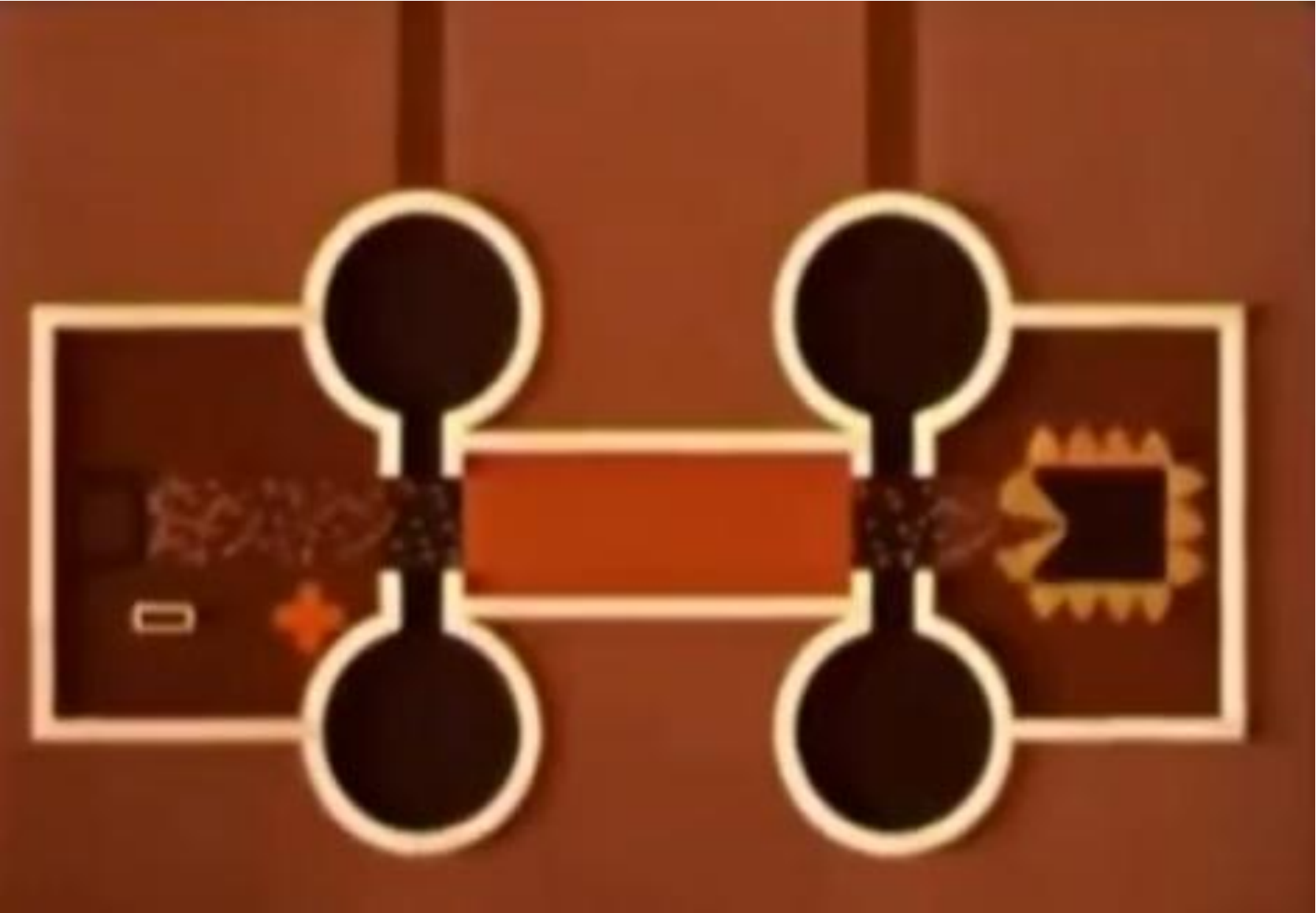
Window / Very low loss material

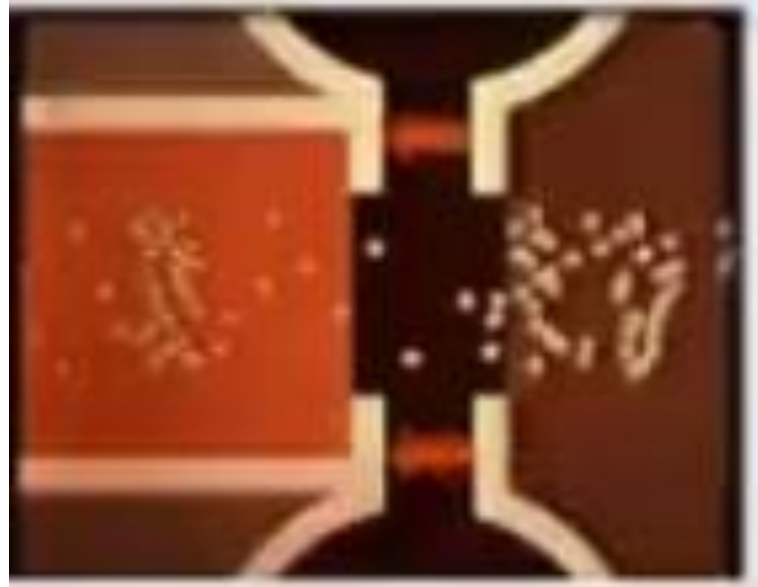
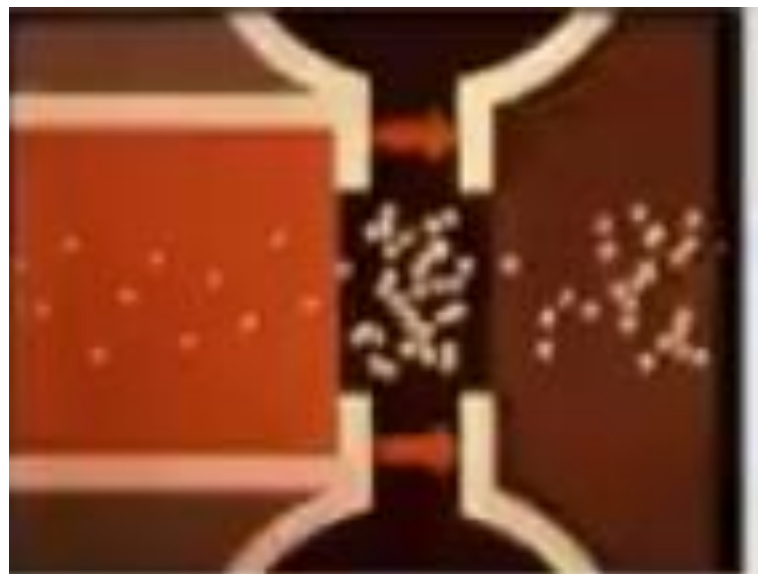
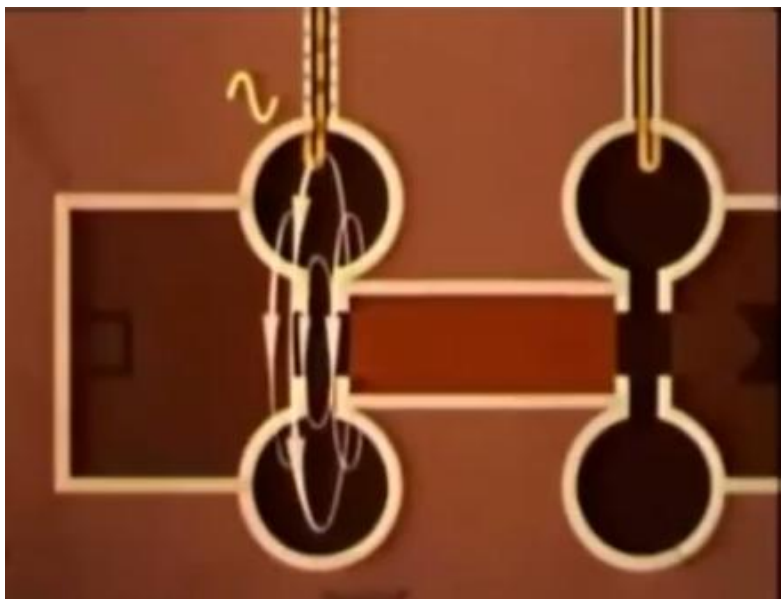


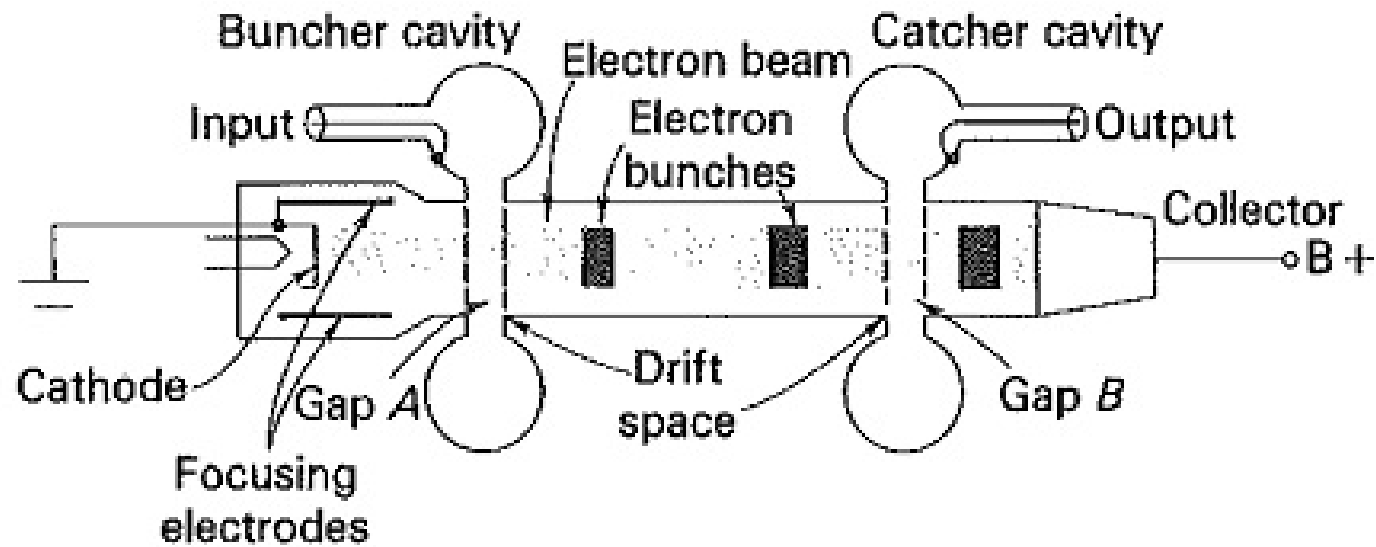
Diamond Window

✓ so that Differential pressure will be sustained ✓
(Ceramic material)









→ Anode

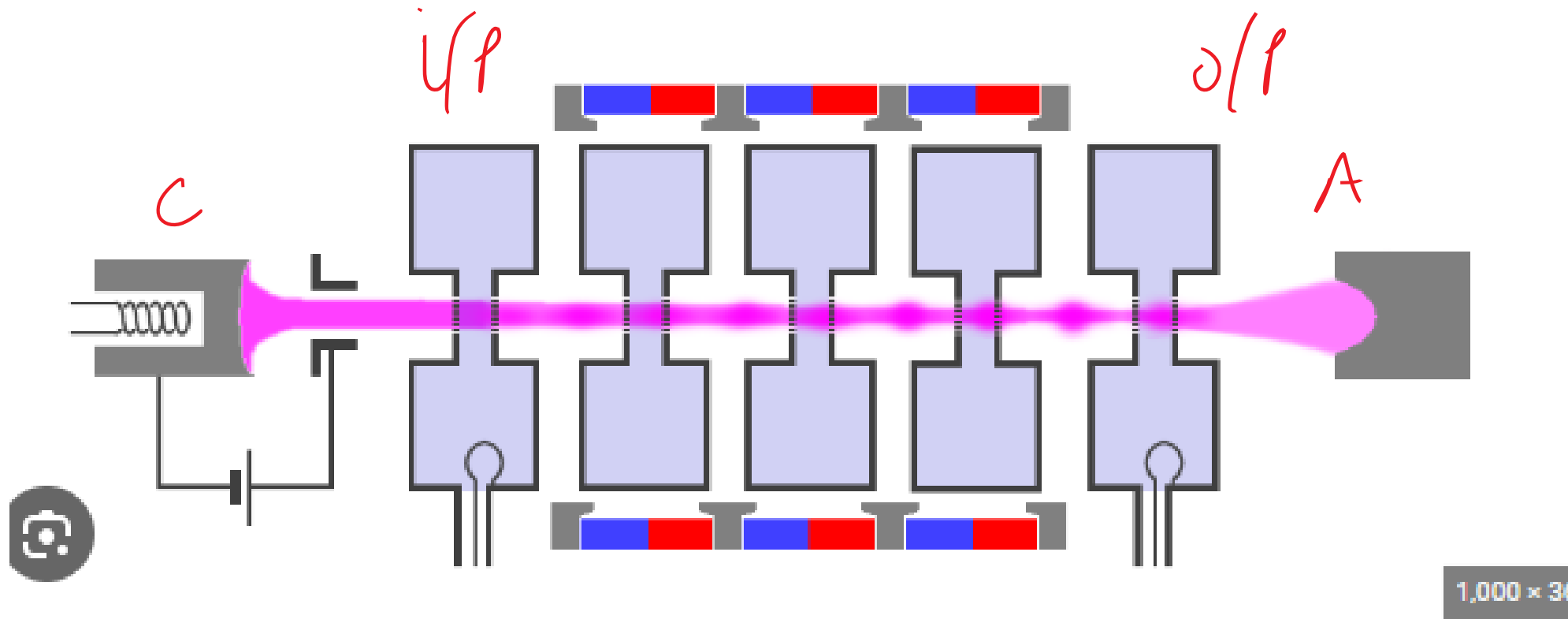
Klystron
image

→ Ceramic
windows
Waveguide

→ Cathode

5 Cavity

Multi Cavity Klystron Resonator



↑ To increase → Amplification
power efficiency

Magnetron
 $\vec{E}$ & $\vec{H}$ are produced

$\perp$ to each other
& crossing each other.

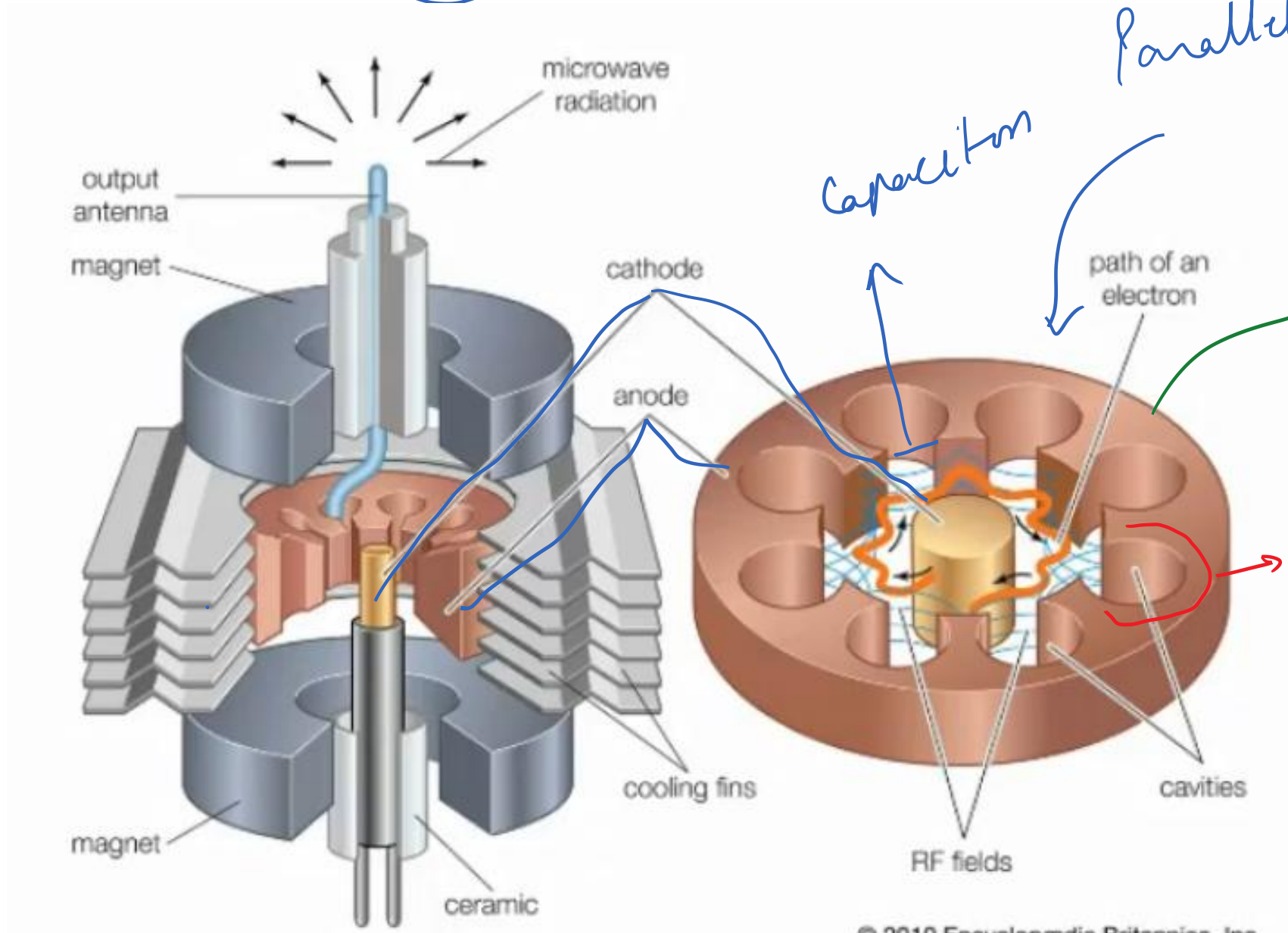
Types of Magnetron!

a) Cyclotron - frequency magnetron
b) negative-resistance magnetron

c) Cavity - Type resonator

Interaction of e^- s with $\vec{E}$ & $\vec{H}$ field.
Principle!

Cylindrical Magnetron (8 cavity Cylindrical Magnetron)



Capacitor

Parallel LC circuit

Resonator

Inductor

* Magnetron! related to Magnets

↓ Use 8 Resonators for getting high power

high freq. signal.

* Magnetron is type of cylindrical diode only.

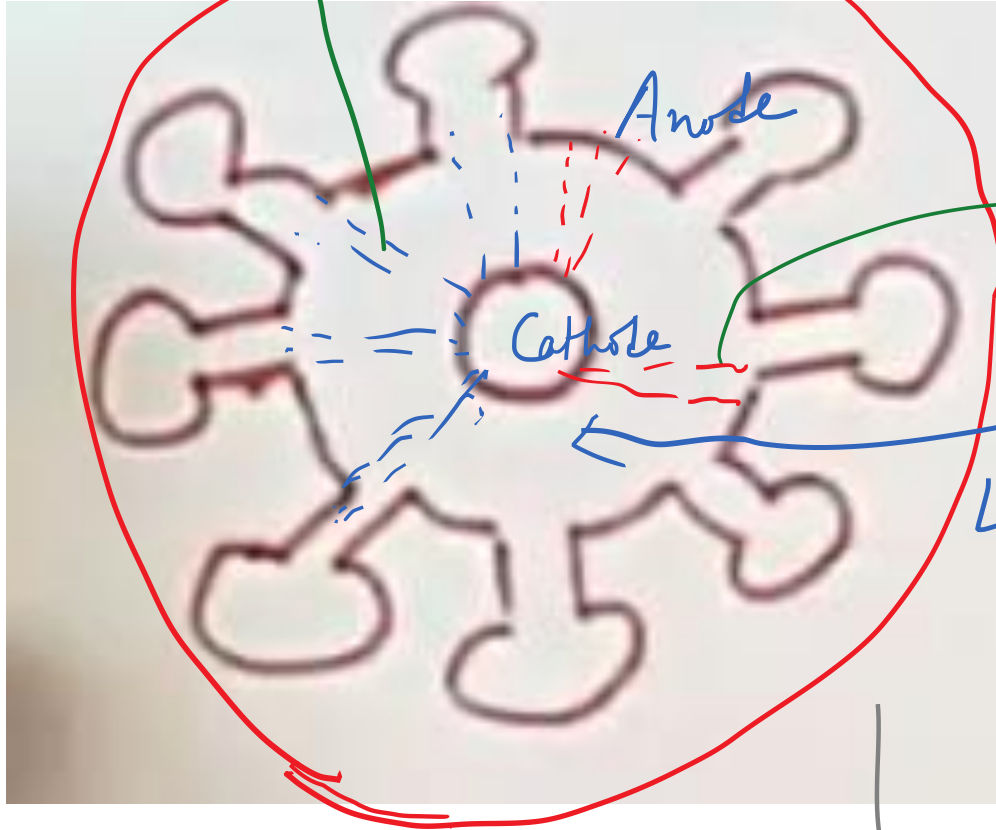
* e^- s will move from cathode to anode

* Not all the energy is being used.

Useful e^- enters into
resonator

Two terminal cathode - anode is
called diode.

[Cylindrical
Diode]

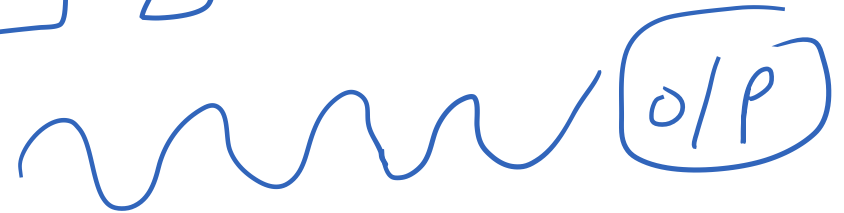


Hits the
wall

putting a potential i/p



Resonant
frequency

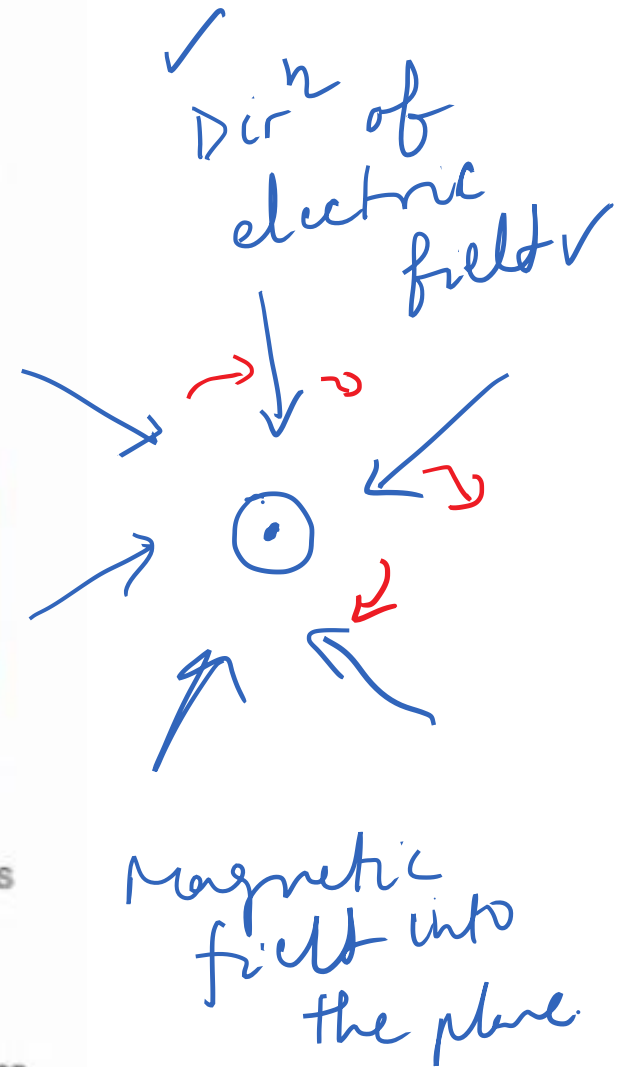
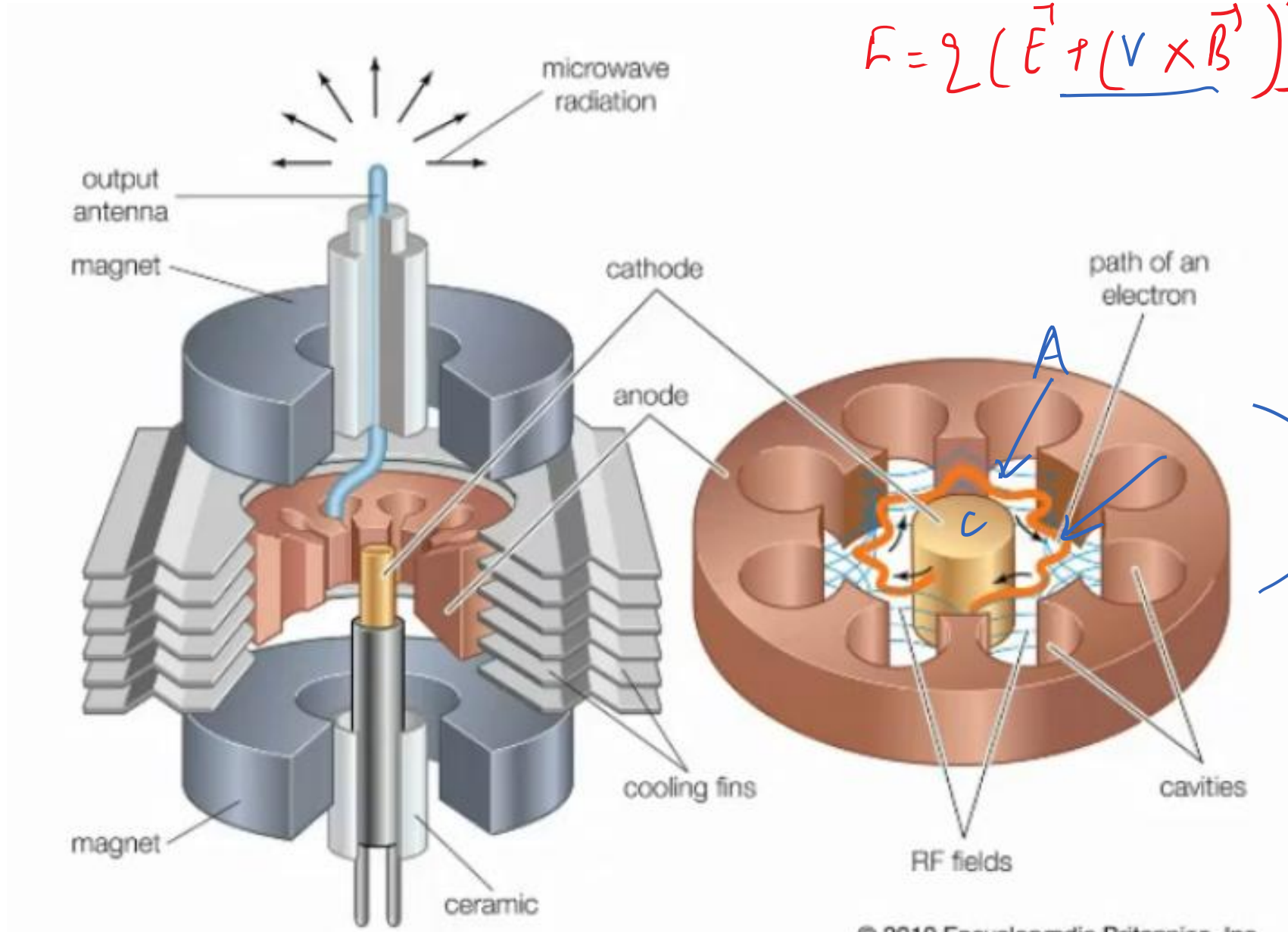


poor efficiency only
3 to 4%.

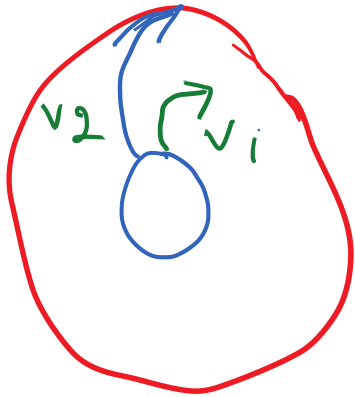
* Modification needed *

Besides potential, Apply the magnetic field as well

$$F = q(\vec{E} + (\vec{v} \times \vec{B}))$$



$$\text{As } \vec{E} = \nabla(\vec{E} + (\vec{v} \times \vec{B}))$$



$$\underline{v_1 > v_2}$$

More $\vec{v}$

more curvature

less $\vec{v}$

More straight path

At steady state, all are at the same potential.

potential.

possible phase shift for adjacent

Transient time resonator

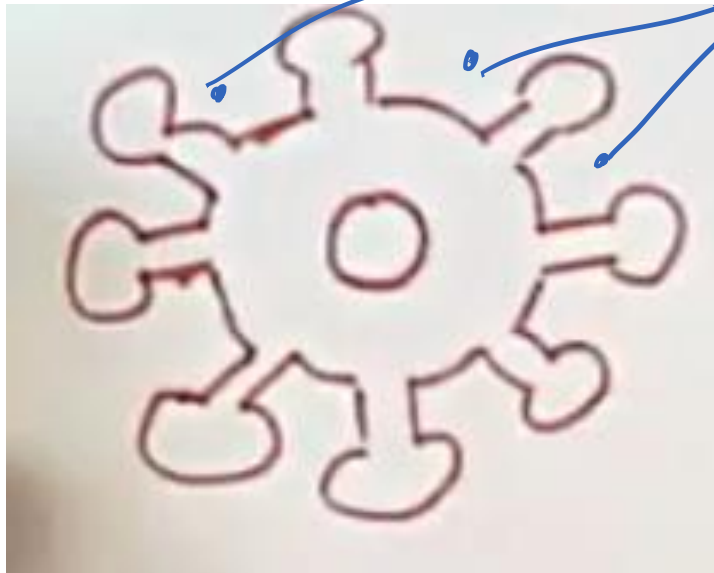
$$n\phi = 2\pi N$$

$$\frac{2\pi}{4\pi} \\ \frac{4\pi}{6\pi}$$

$$\boxed{\phi = 45^\circ N}$$

$$\text{So } \underline{\phi = 45^\circ}$$

adjacent resonator phase shift.
90, 135, 180



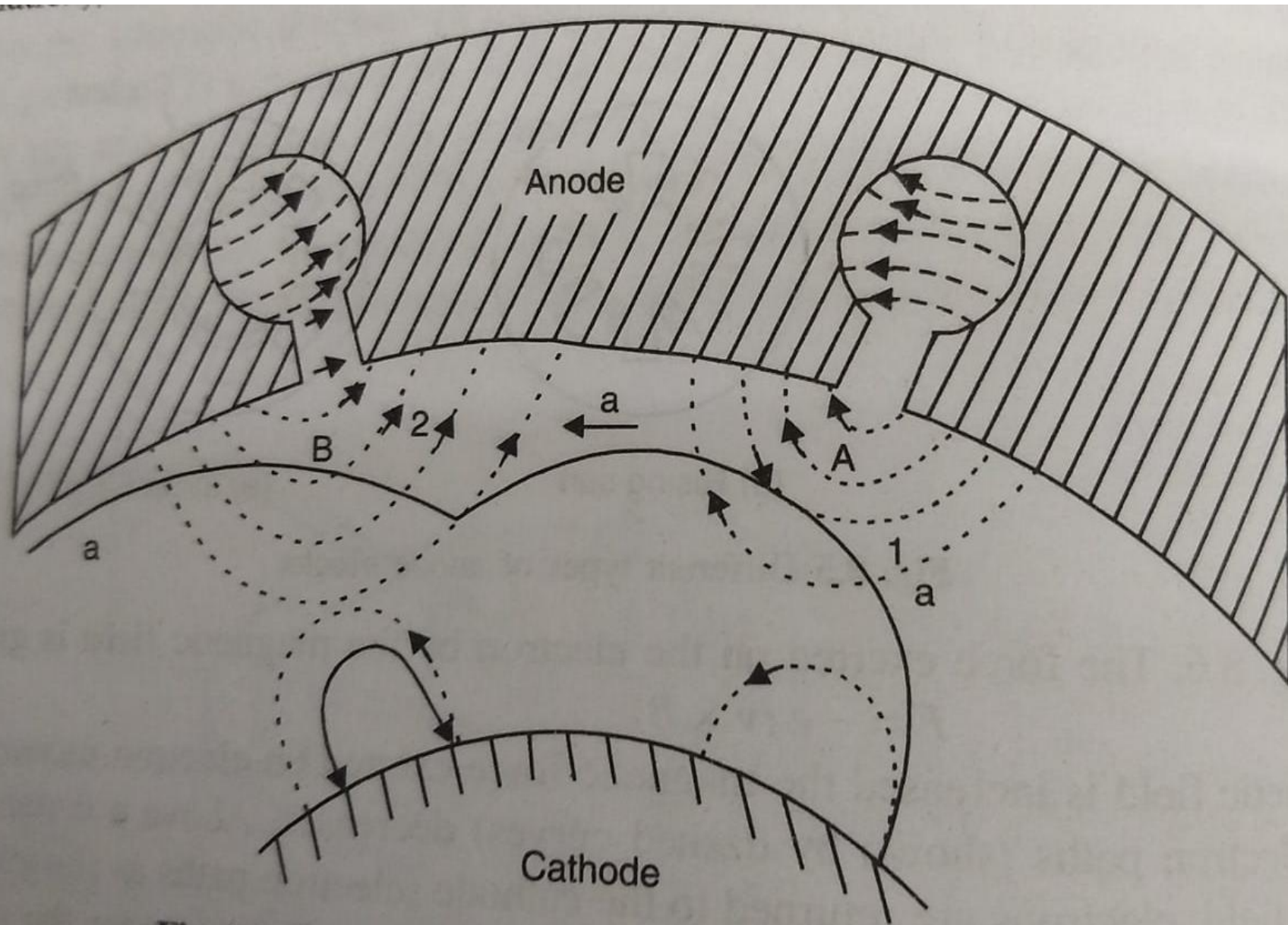
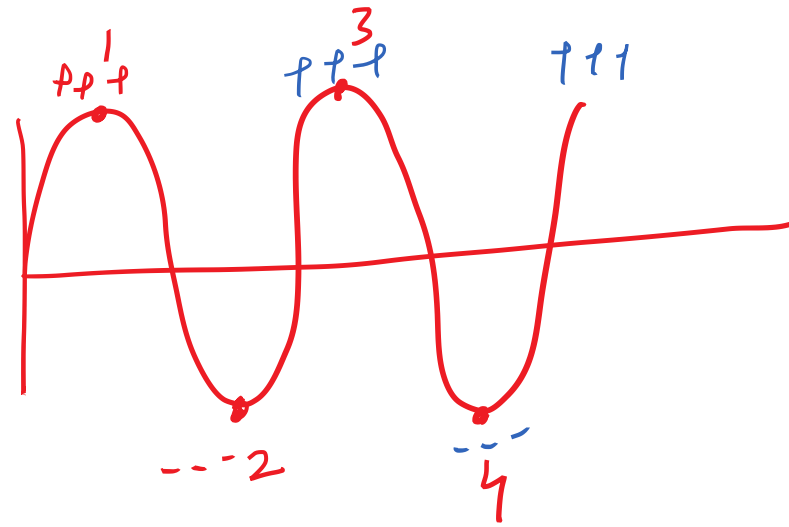
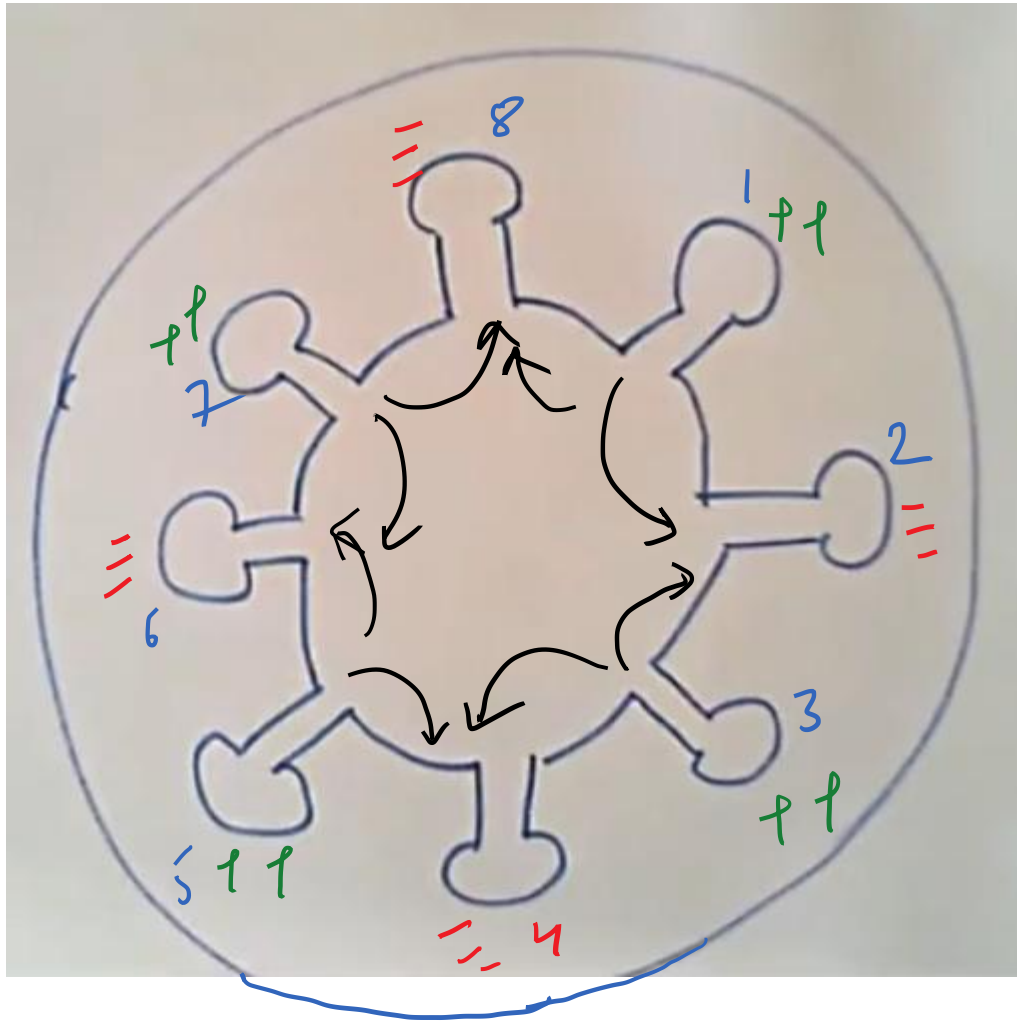


Fig. 8.7 Electron trajectories in the presence of r.f. field

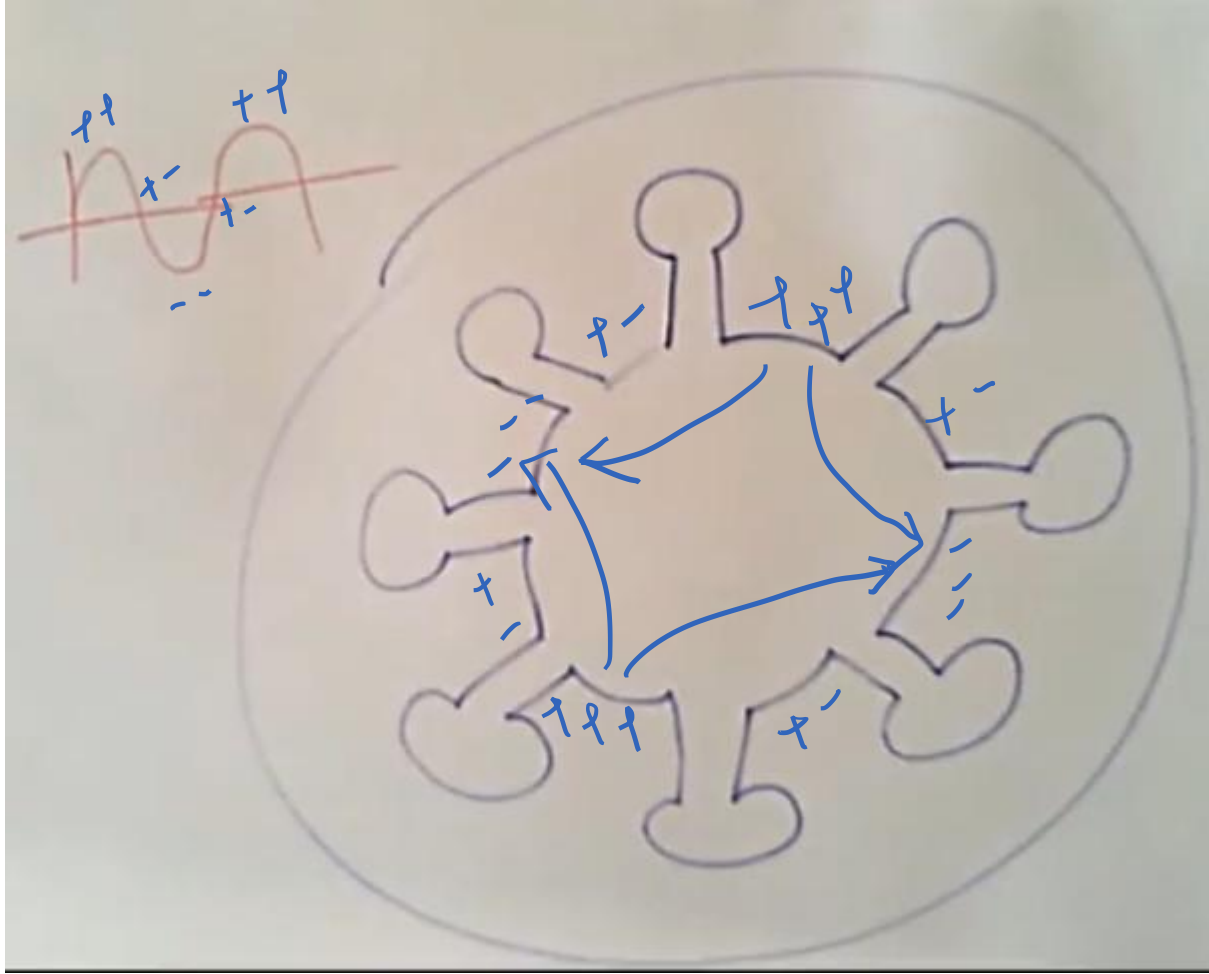
Case 1. When the phase shift in adjacent one is 180° .



Transient
time

when they are 180°
apart.

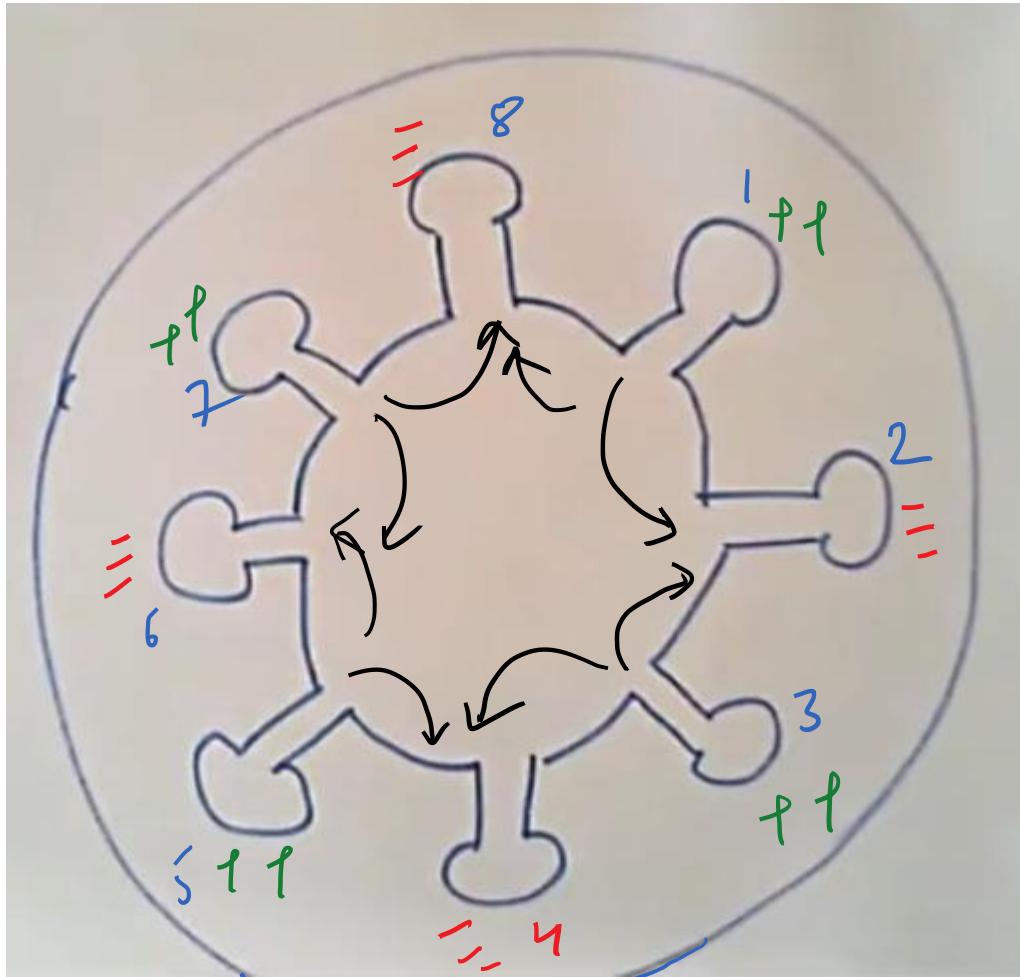
when the resonators are π apart.



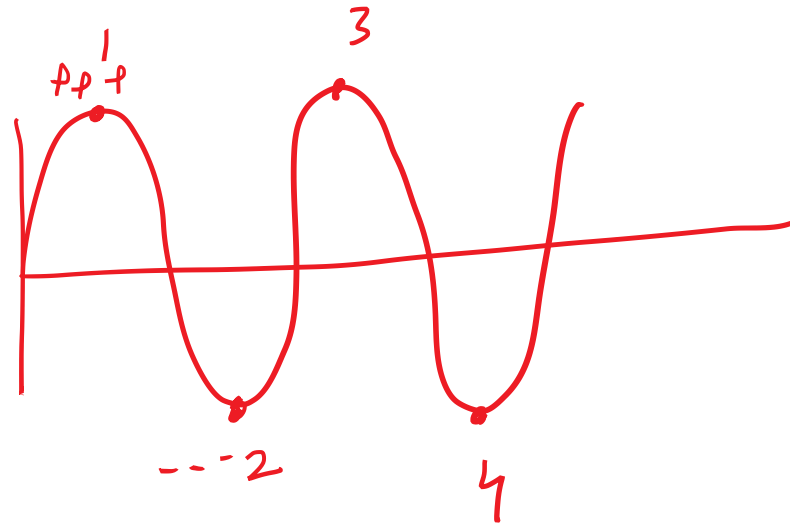
The cavity can support multiple field configurations. Each field is corresponding to a mode & frequency.

π -mode is more important one

Case 1. When the phase shift in adjacent one is 180° .



patterns happen in

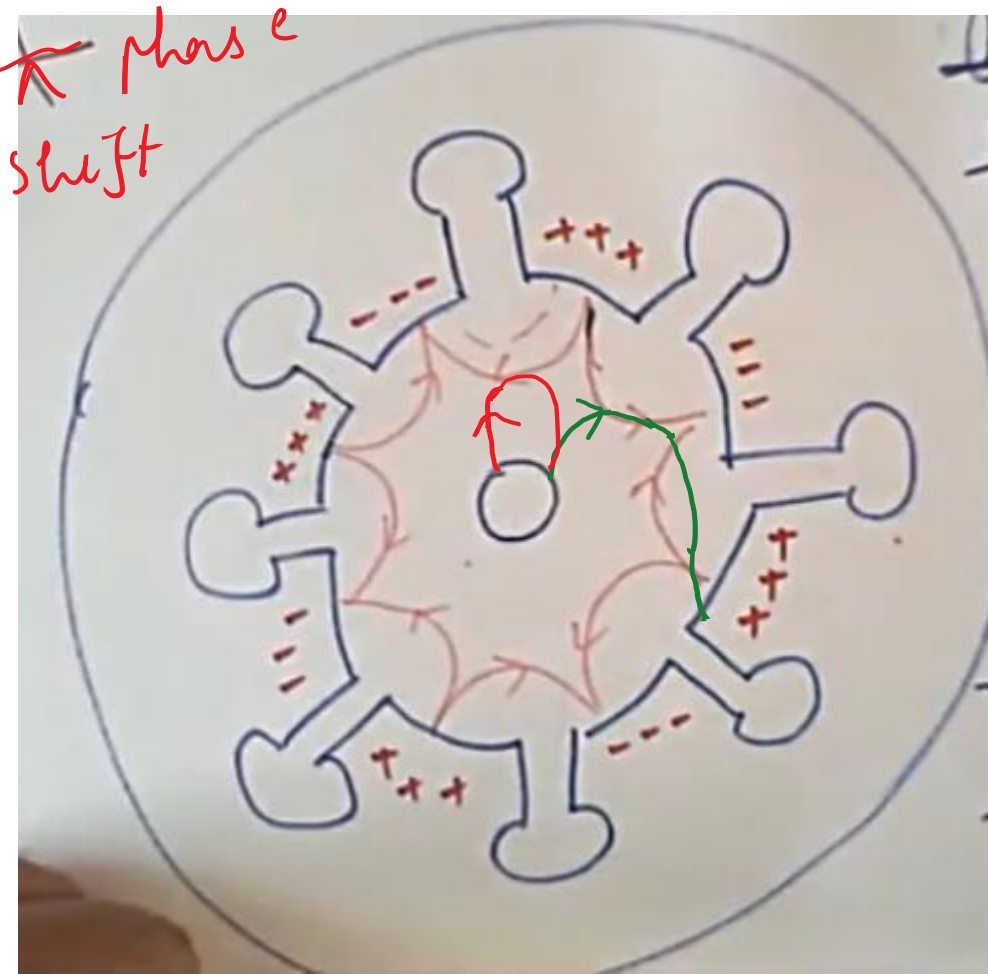


Transient
time

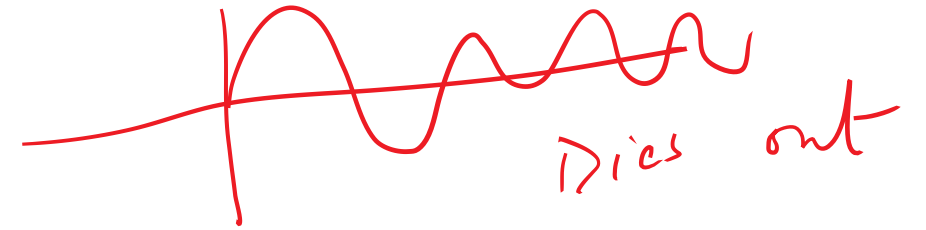
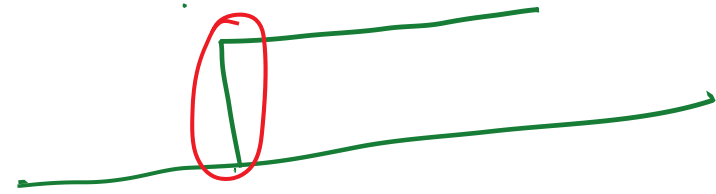
when they are 180°
apart.

transient mode, after that
they die out. What to do??

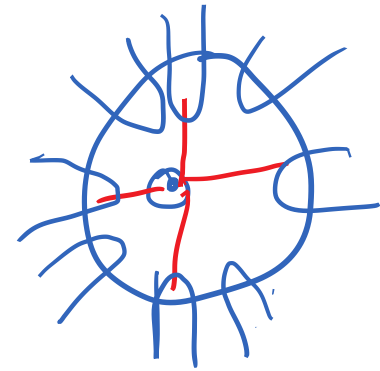
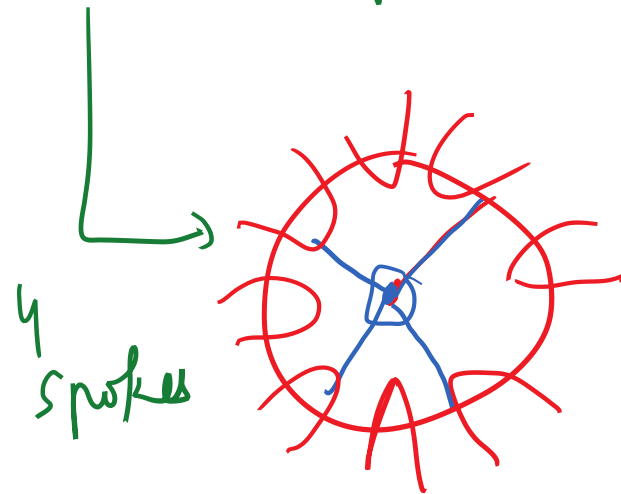
8 Cavity Magnetron



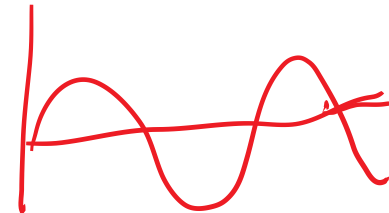
~ cavity



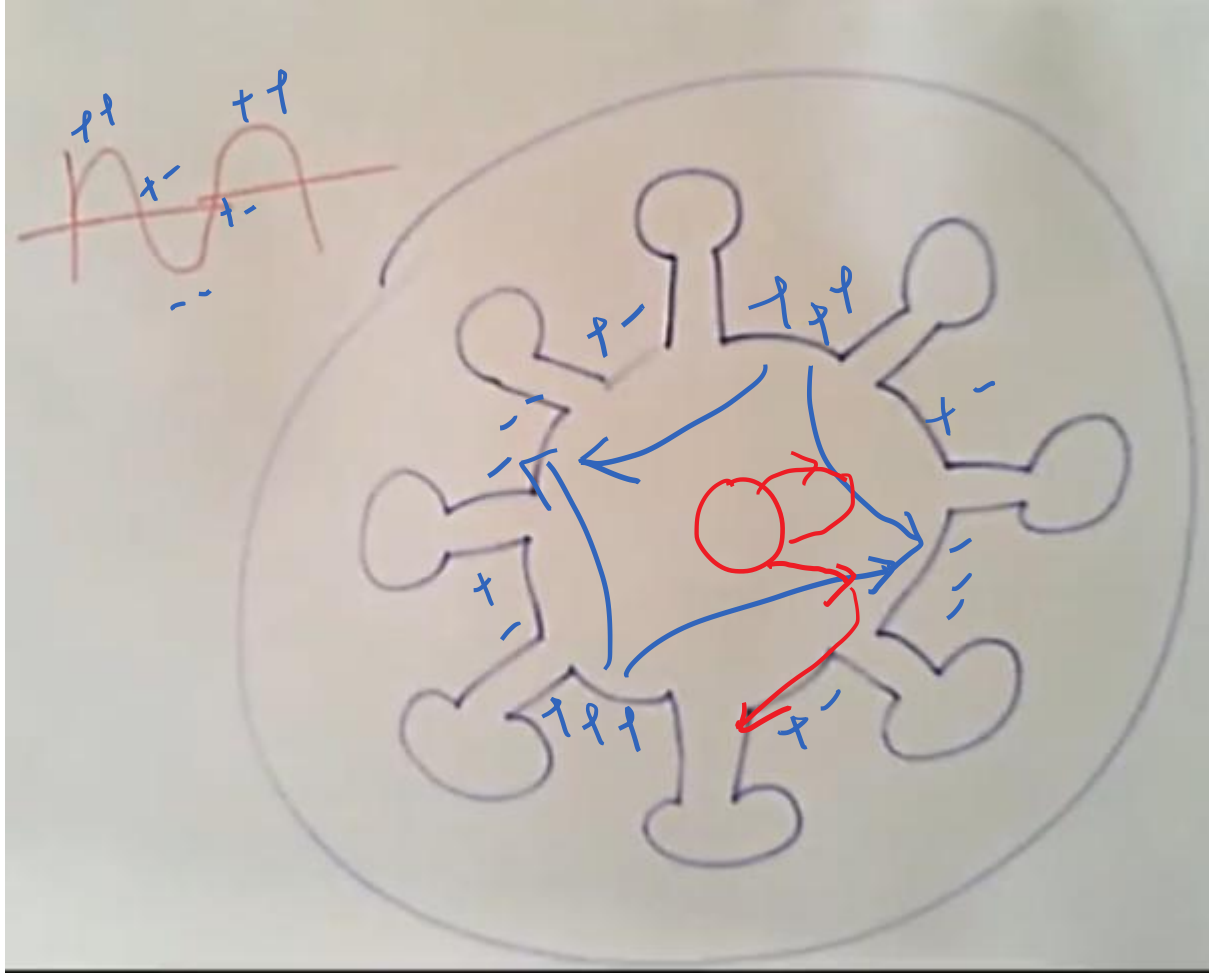
8 Cavity



$N/2$ spokes



when the resonators are π apart.



So only 2 spokes

* * *

So π phase shift

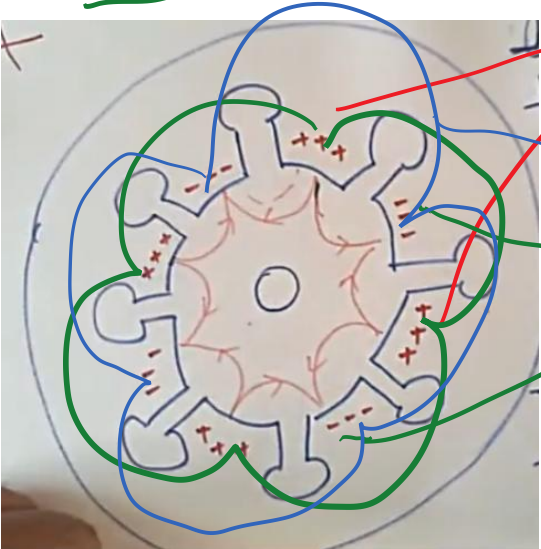
Cavity resonators covers
all the 8 cavity using
4 - 4 spokes alternatively.

* Different field patterns / Different modes / Different frequencies.

Mode jumping in Magnetron

* Most preferred mode is π mode.

Solⁿ



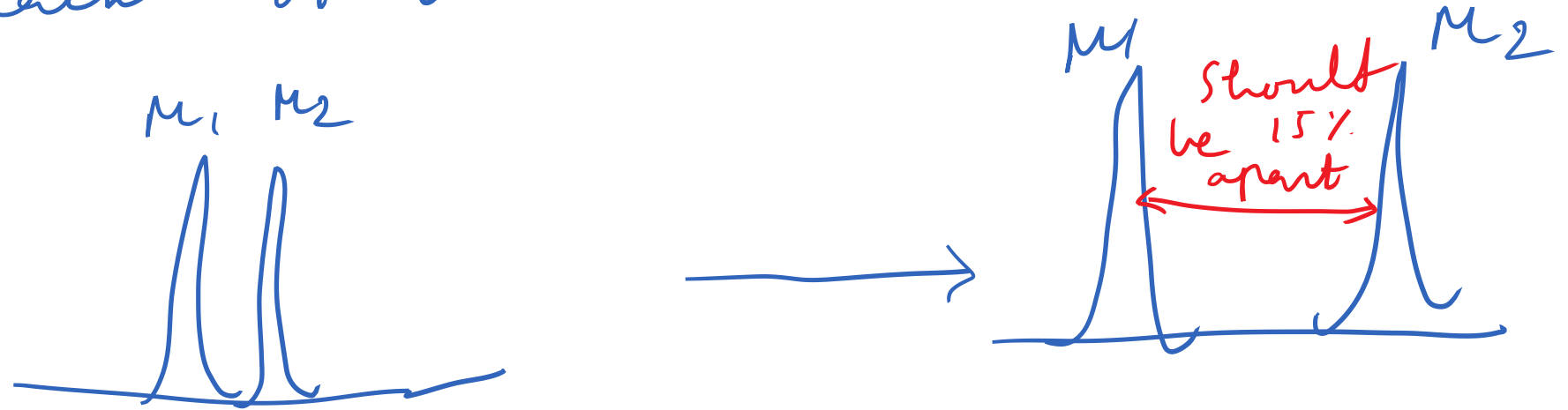
Alternate resonators are at same potential.

→ Strapping the magnetron

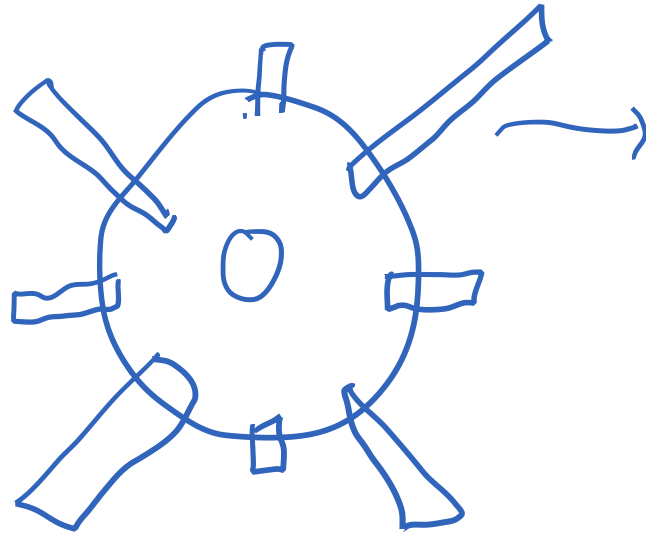
Alternate resonators are at same potential.

So that modes will not jump.

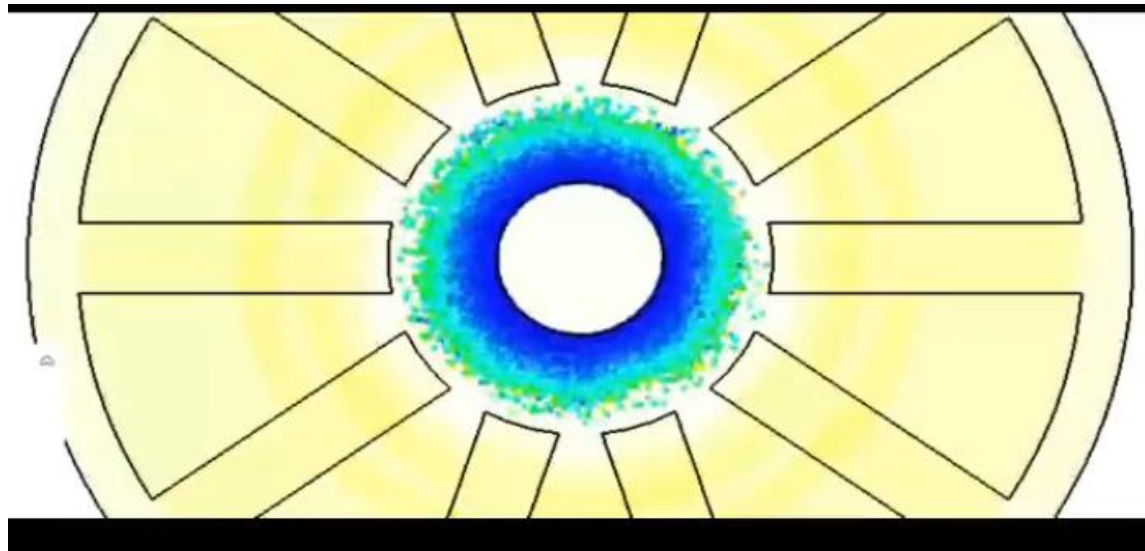
* There is other possibility that modes can be
close to each other.



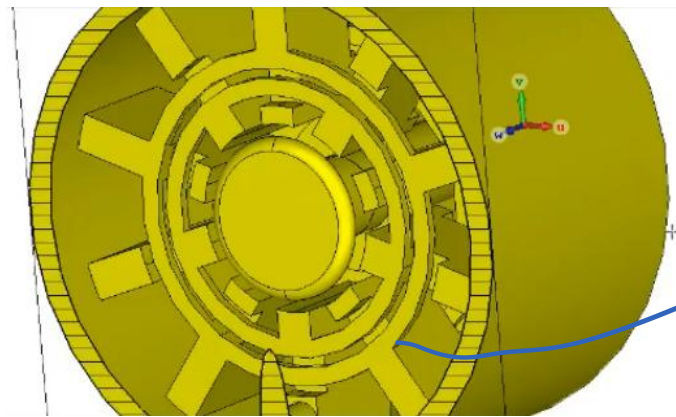
Rising Sun
magnetron



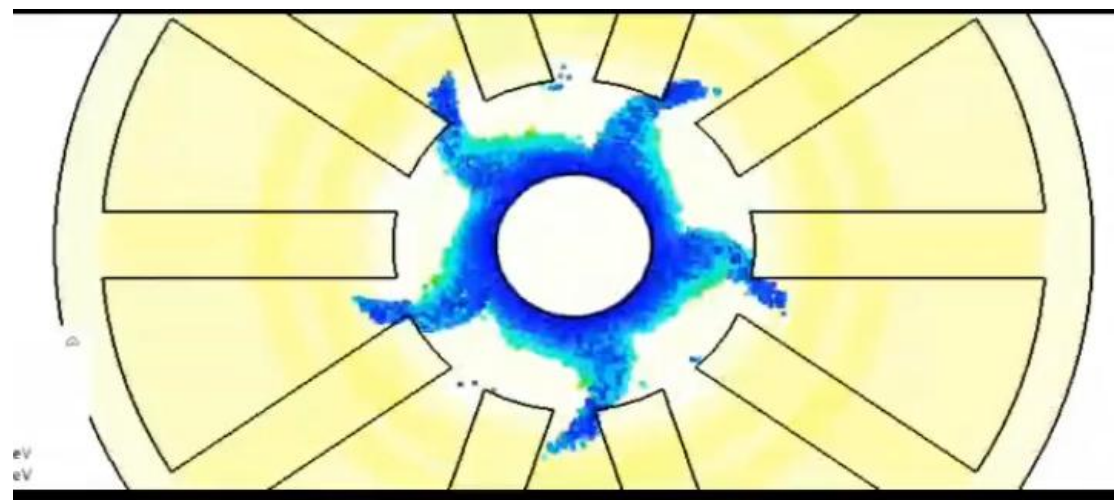
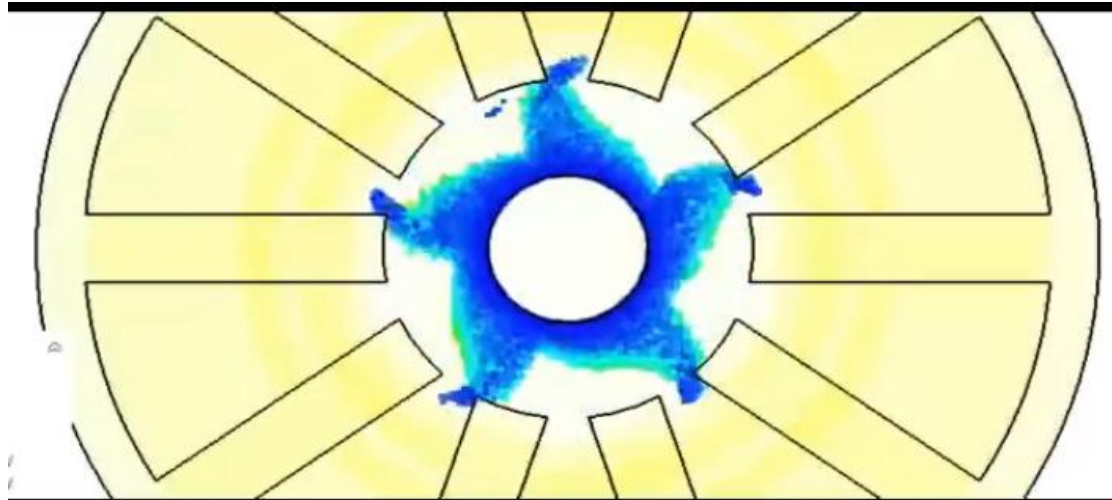
Different sizes

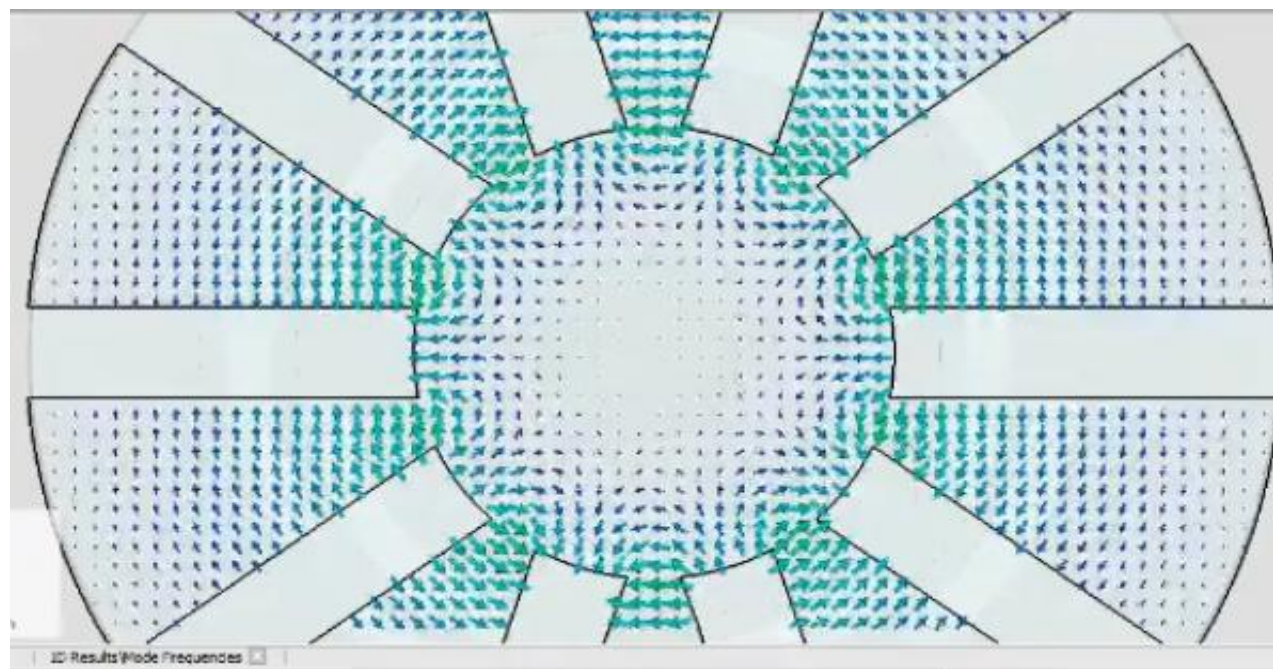
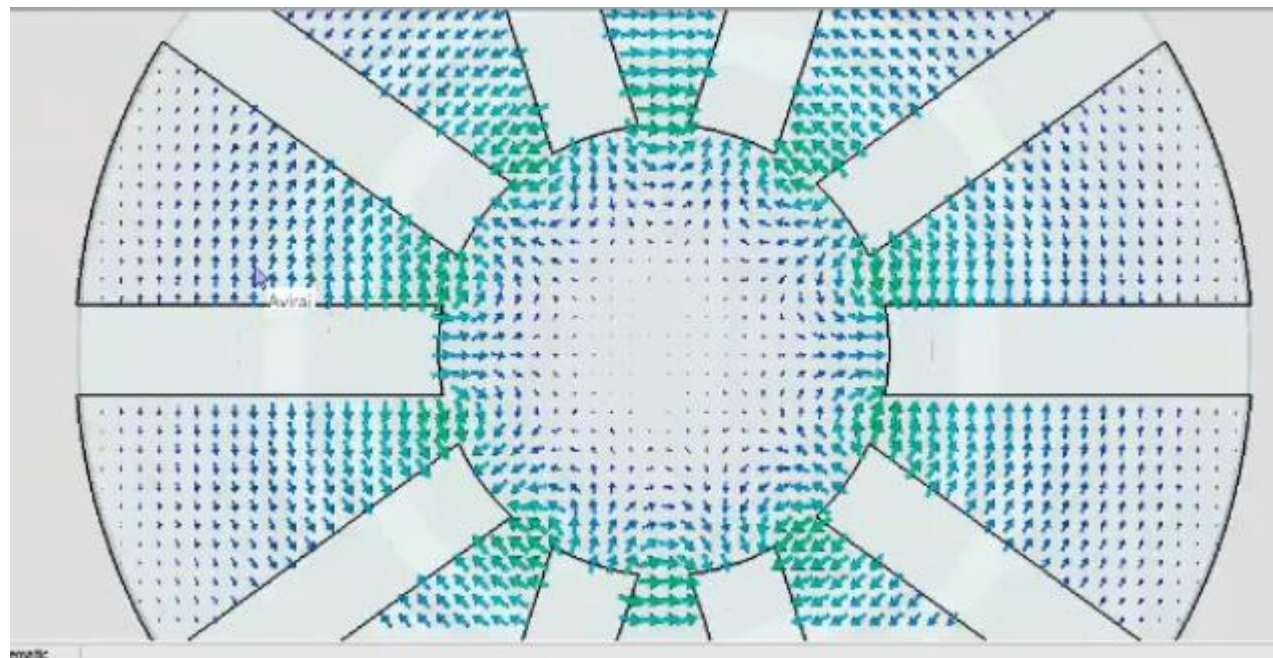
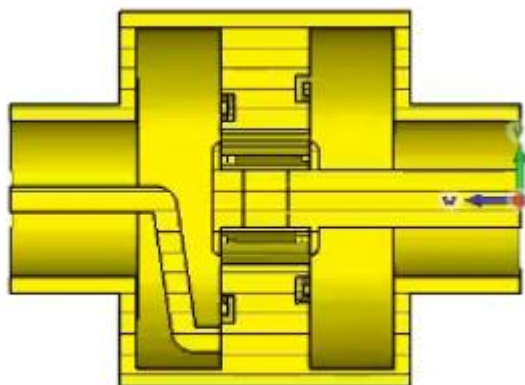
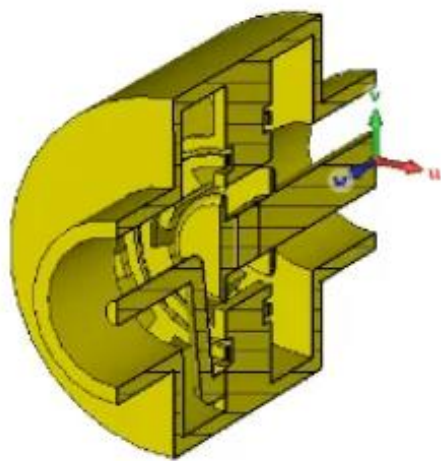
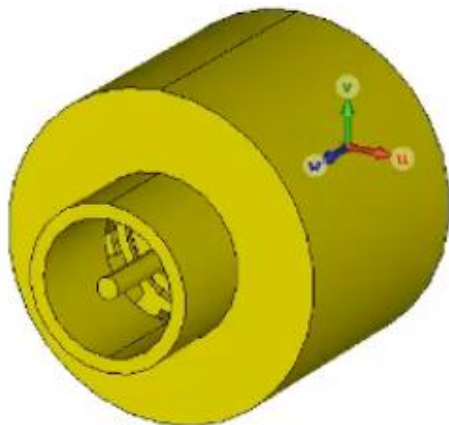


10 Cavity Magnetron

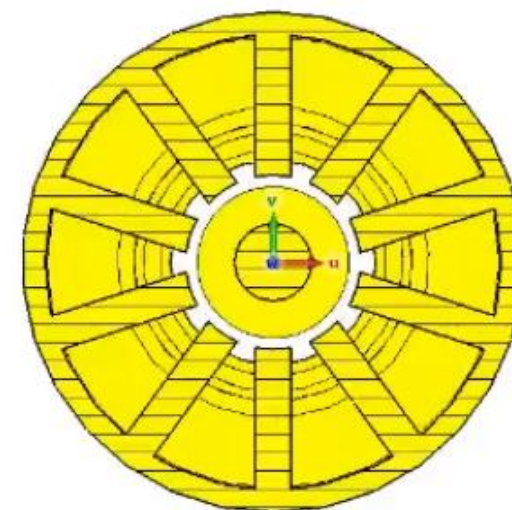


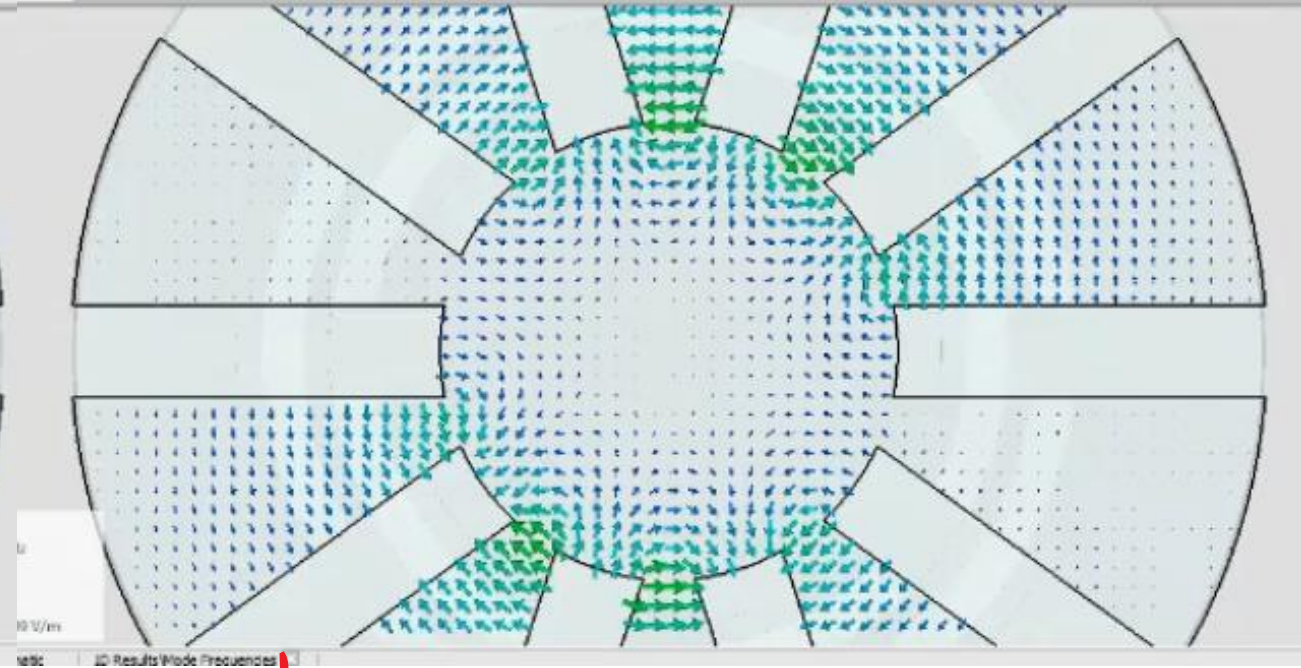
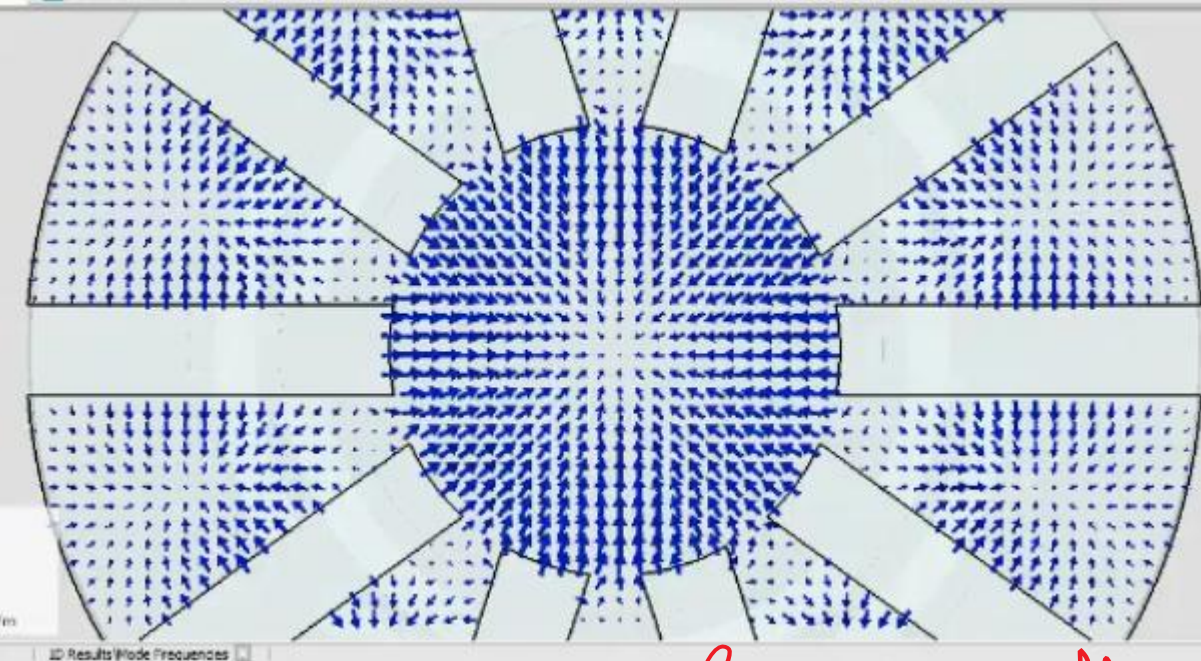
Trapping
of
magnetron



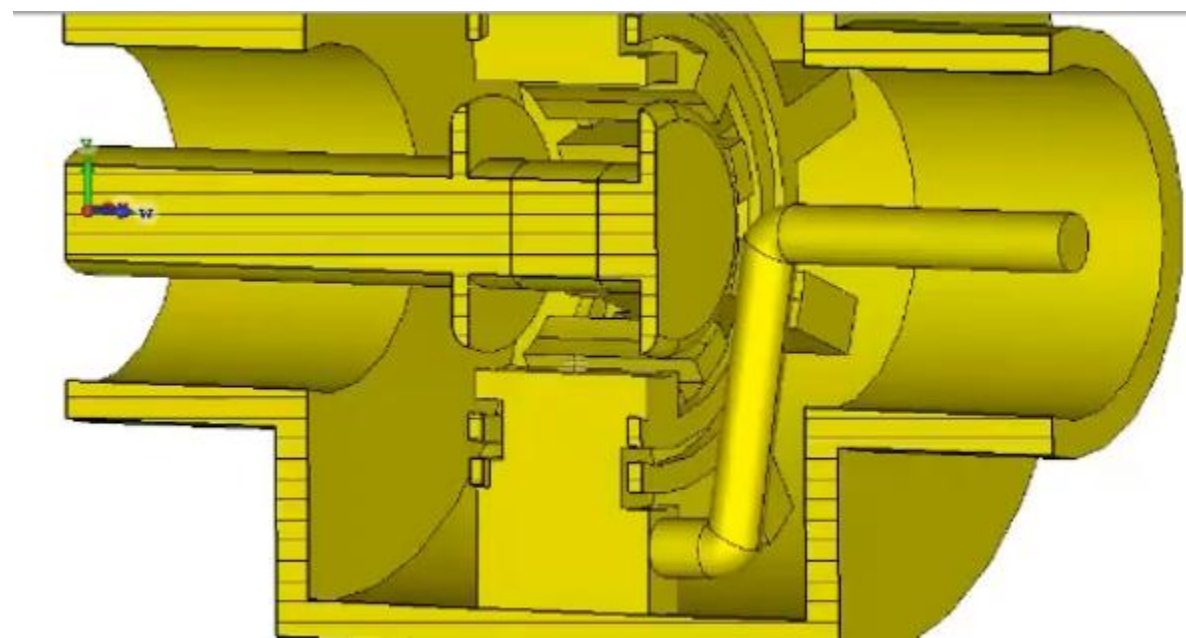
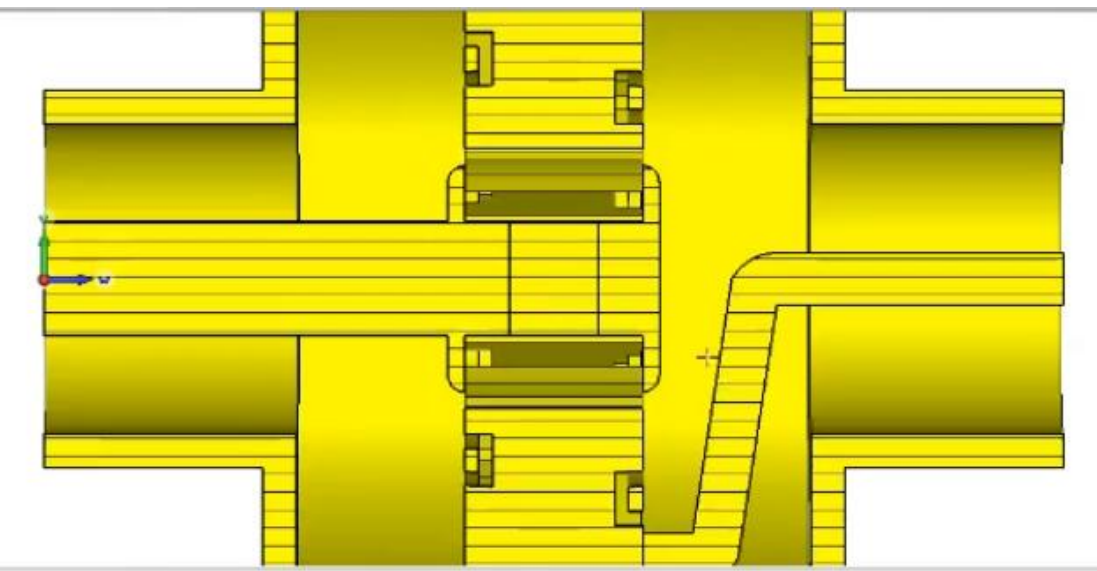


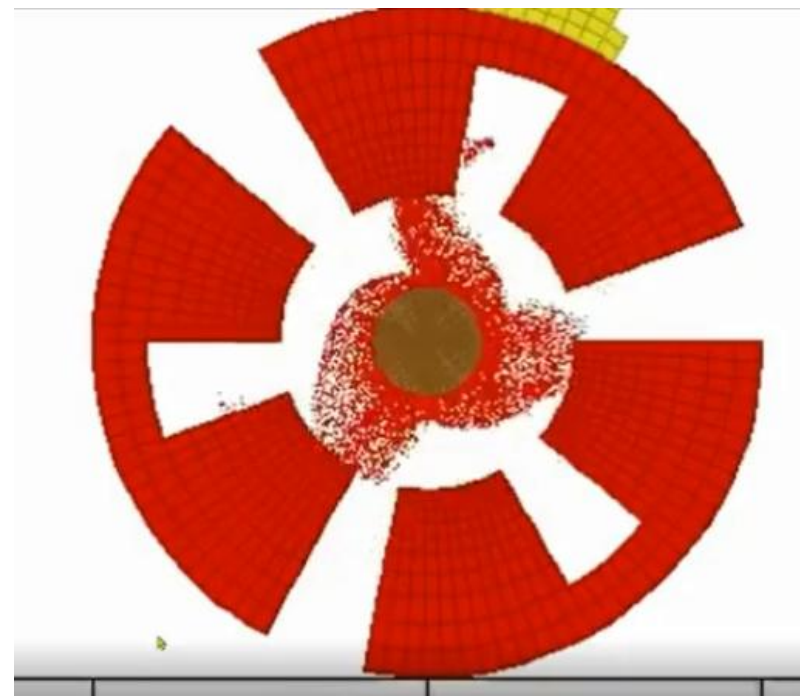
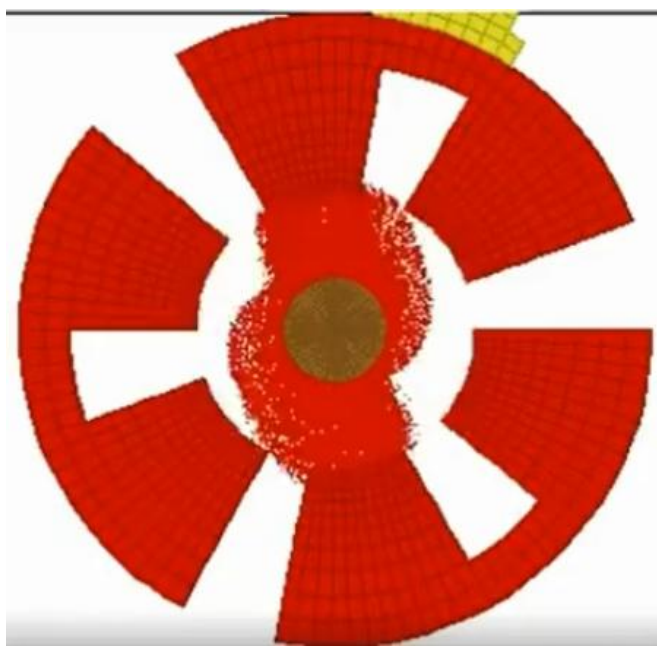
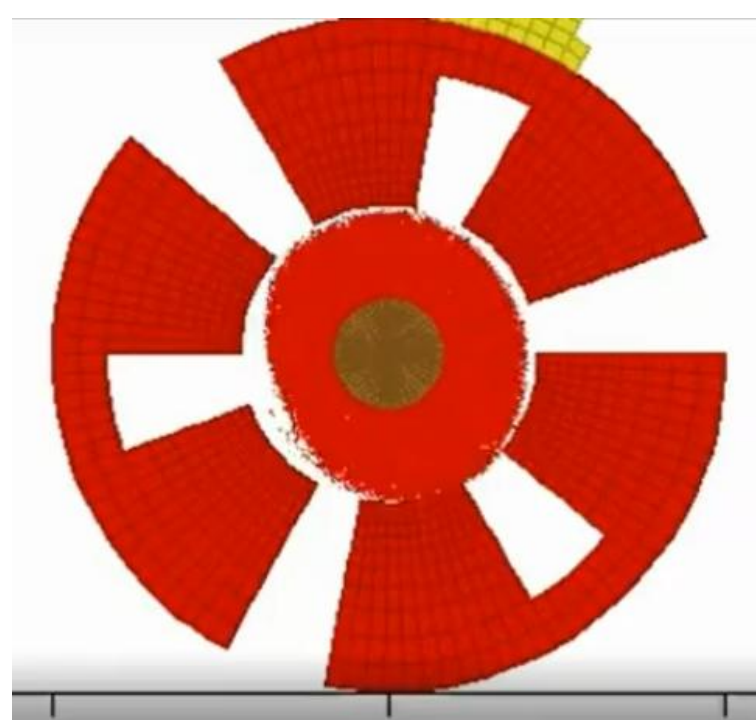
π Mode
for
Magnetron



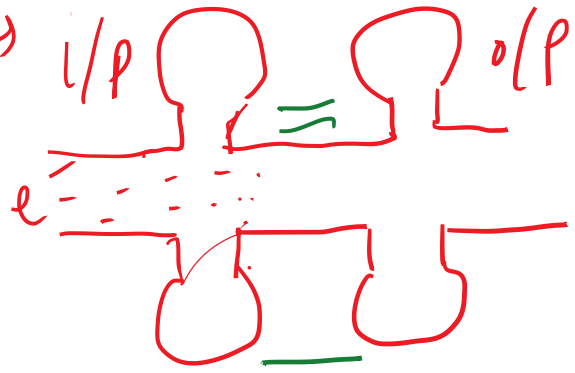


Some other modes





Klystron →
Magnetron →



→ Magnets
to converge the
electric field

Amplifier
High power source → Work at
high
frequency

→ Magnetic field plays an
active role to generate the
signal

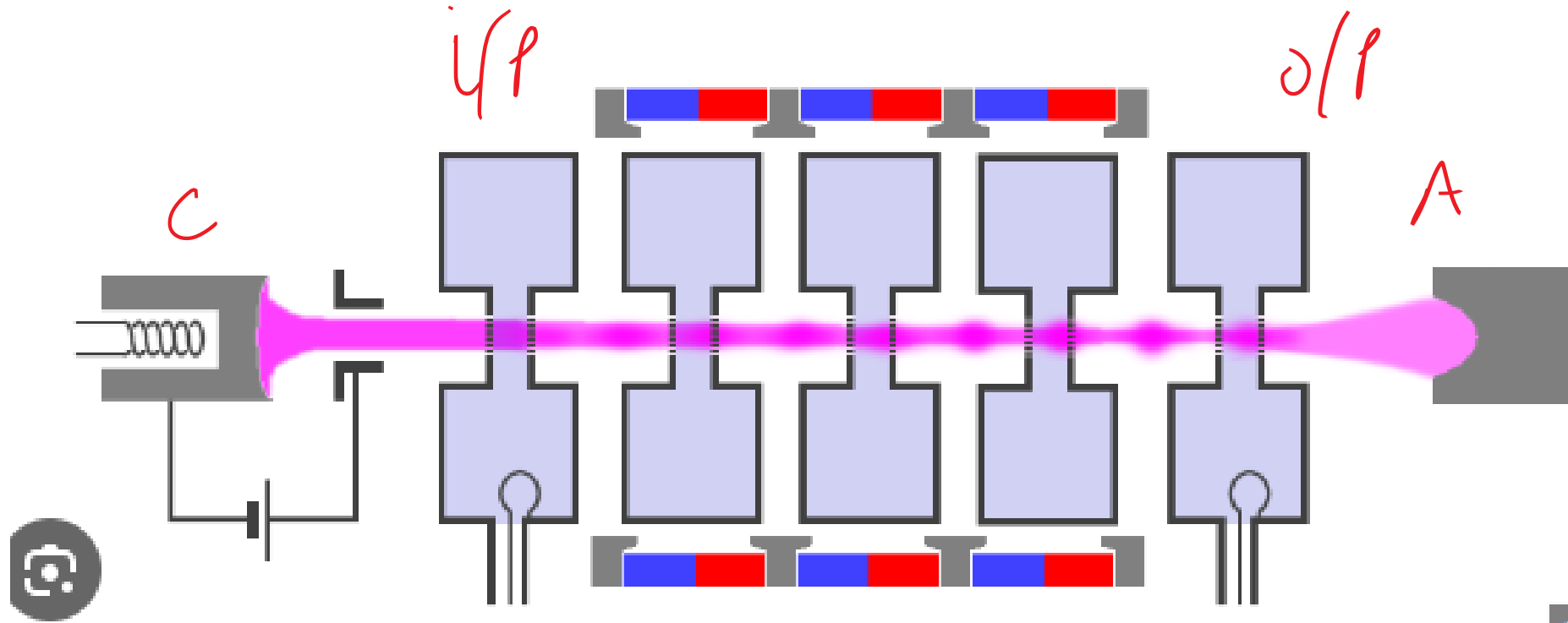
→ M-type tube

→ Magnetic field for
focussing
→ O-type tube

* How to increase the gain
in Klystron
✓ By more bunching of e^- s
✓ By more interaction

5 Cavity

Multi Cavity Klystron Resonator



* e^- velocity
 $\rightarrow 1.5 \text{ m/s}$

* field velocity
 $-3 \times 10^8 \text{ m/s}$

1,000 x 34

$\uparrow$ to increase $\rightarrow$ Amplification power efficiency

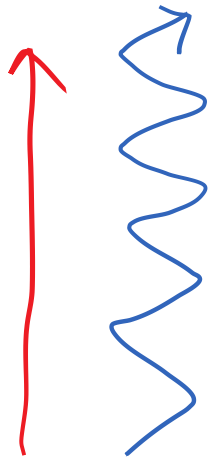
Surfing example

either ask e^- s to move faster $\rightarrow$ Not possible

ask E field to go slower

$$v = \frac{1}{\sqrt{\mu\epsilon}}$$

$\hookrightarrow$ So go with
reducing velocity



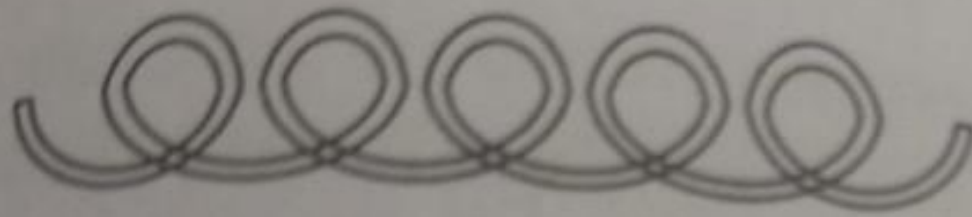
longer path
will lead to

effective axial
less velocity for field

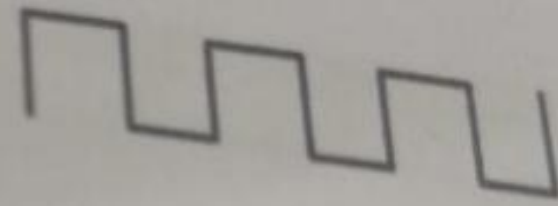
$$\hookrightarrow \underline{10^5 - 10^6 \text{ m/s}}$$

e^- - E field interaction is more

leads to more gain for Amplification.



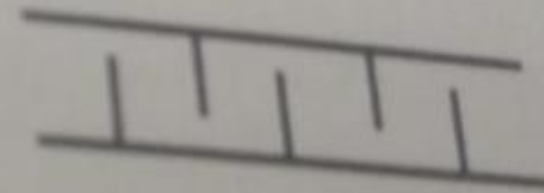
(a)



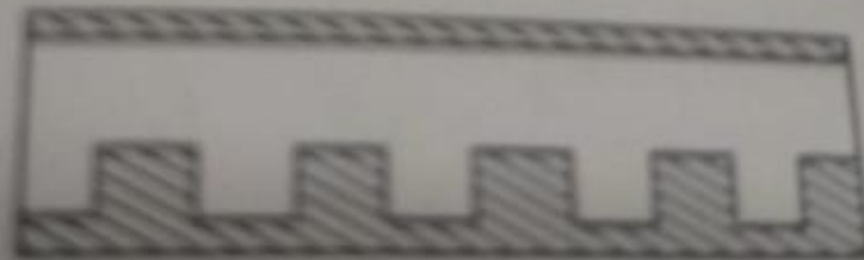
(b)



(c)



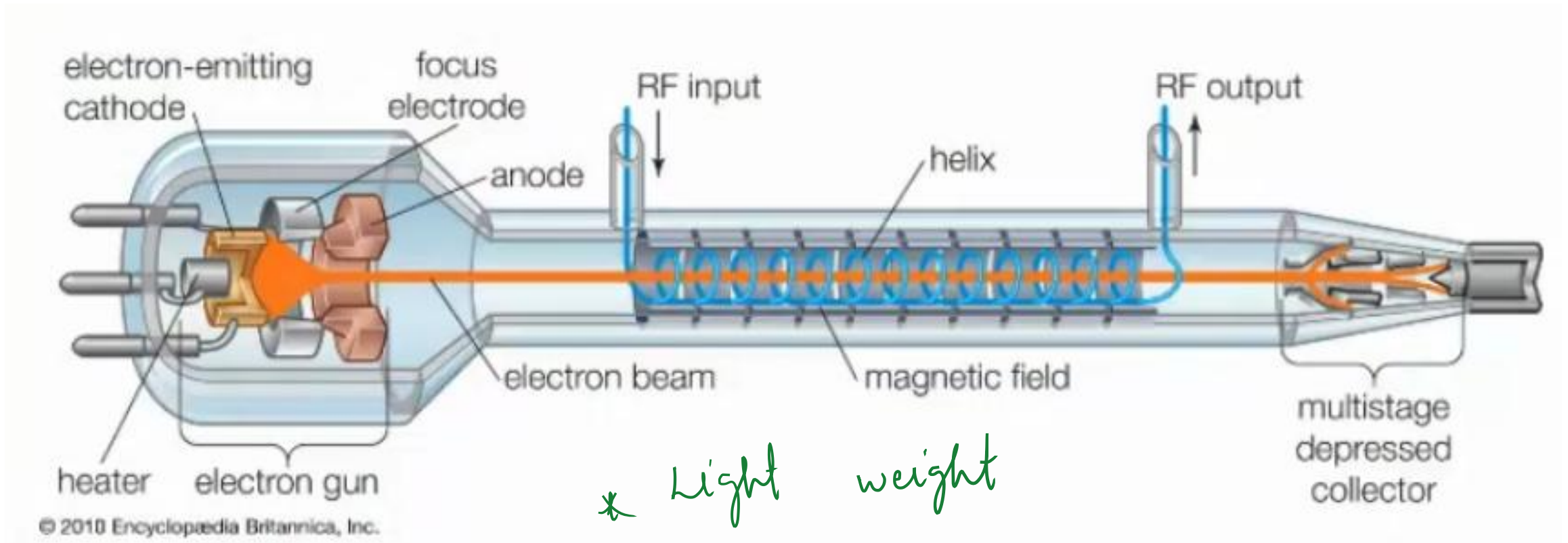
(d)



(e)

Travelling Wave Tube

Slow wave structures



* Light weight

* Space applications
* More Bandwidth

5G High Power Amplifier
6G

TWT

- ① RF signal travels along the beam
- ② Interaction b/w e^- & E field is continuous
- ③ Non-resonant type.
- ④ High power o/p
- ⑤ Helical type structure.

Klystron

- ① RF signal is stationary & only the beam travels.
- ② Occurs only at the gap of cavity
- ③ Resonant type
- ④ Low power o/p comparatively
- ⑤ Buncher & catcher cavity.

① A 250 kW pulsed cylindrical magnetron is operated with the following parameters:

Anode voltage = 25 kV

Peak anode current = 25 A

Magnetic field $B = 0.35 \text{ Wb/m}^2$

Determine a) The efficiency of magnetron
b) The cyclotron freq. ω_c

solⁿ

a) $\eta = \frac{P_{out} \times 100}{P_{in}}$

$$= \frac{250 \times 10^3}{25 \times 10^3 \times 25} \times 100 = 40\%$$

b)

$$\frac{m v^2}{2} = \sum B$$

$$\frac{m h \omega}{h} = \sum B$$

$$\omega = \frac{\sum B}{m}$$

$$\omega = \frac{0.35 \times 1.6 \times 10^{-19}}{9.11 \times 10^{-31}} = \underline{0.616 \times 10^{11} \text{ rad/sec}}$$

A linear magnetron has the following operating parameters: operating freq = 8 GHz

Magnetic flux density $B = 0.015 \text{ Wb/m}^2$

Distance b/w anode and cathode

$$d = 5 \text{ cm}$$

stub surface radius = 2.77 cm

Calculate the electron velocity at hub surface.

Solⁿ

$$\frac{mv^2}{\hbar} = \sum V B$$

$$\frac{m\hbar\omega^2}{\hbar} = \sum V B$$

$$m\hbar \frac{v^2}{\hbar^2} = \sum V B$$

$$v = \frac{\sum B \hbar}{m}$$

$$v = \frac{9.6 \times 10^{-19}}{9.11 \times 10^{-31}} \times 0.015 \times 2.77 \times 10^{-2} = \underline{0.73 \times 10^3 \text{ m/s}}$$

$$v = \hbar \omega$$